

# Line spectra

## Learning outcomes

By the end of this section you should be able to:

- Qualitatively describe the relationship between colour, wavelength, frequency and energy across the electromagnetic spectrum.
- Distinguish between a continuous and a line spectrum.

## Elements from the Sun

The element helium (symbol He) was first discovered in 1868 by a French astronomer named Pierre Jules César Janssen. While viewing a solar eclipse through a prism\*, he observed a bright yellow line originating from the Sun's chromosphere. The element was named helium after the Greek word 'helios' meaning Sun. The study of the interaction between matter and light is called spectroscopy. Using this technique, elements can be identified by observing their unique emission spectra just like people can be identified by their fingerprints. In this section, we will learn how these emission spectra are produced in terms of electron transitions between energy levels in the atom.

\* Be aware that you should never look directly at the Sun; doing so might cause serious eye damage.

## Creativity, activity, service

**Strand:** Service

**Learning outcomes:**

- Demonstrate how to initiate and plan a CAS experience
- Demonstrate engagement with issues of global significance

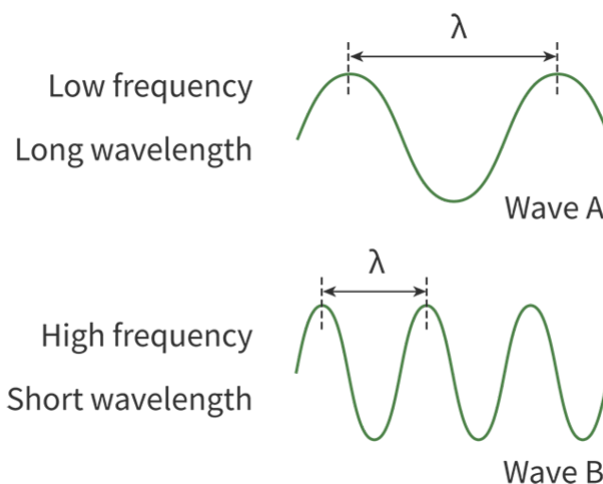
The Sun is the source of all of the Earth's natural light. Not only does it provide the necessary light for plants to grow, it can also be used as an energy source in the form of solar power  (<https://www.energy.gov/eere/solar/how-does-solar->

work). One important utilisation of solar power is the solar oven (<https://www.treehugger.com/what-are-solar-ovens-5088602>), a device that uses heat and light from the Sun to cook food. According to Solarcookers.org (<https://www.solarcookers.org/>), approximately one in three people lack sufficient fuel to cook food.

A possible CAS experience could be to raise awareness of the use of solar power, particularly among countries where people lack access to sufficient fuel for cooking.

## The electromagnetic spectrum

When you think about light, what first comes to mind? It is probably the visible light that allows us to see. However, visible light is just a small part of what is known as the electromagnetic spectrum, or the EM spectrum (**Figure 2**). The EM spectrum is divided into seven regions arranged in order of frequency, wavelength or energy. The wavelength,  $\lambda$ , is the distance between two crests in an oscillating wave. It has units of distance (m). The frequency,  $f$ , is the number of waves that pass a point in one second. It has the units hertz (Hz) or  $\text{s}^{-1}$ . All regions of the EM spectrum travel at the same speed – the speed of light, which is  $3 \times 10^8 \text{ m s}^{-1}$ . Frequency and wavelength are inversely proportional, meaning that if frequency increases, the wavelength decreases. **Figure 1** shows two waves – wave A has a longer wavelength and lower frequency than wave B.



**Figure 1.** The relationship between wavelength and frequency.

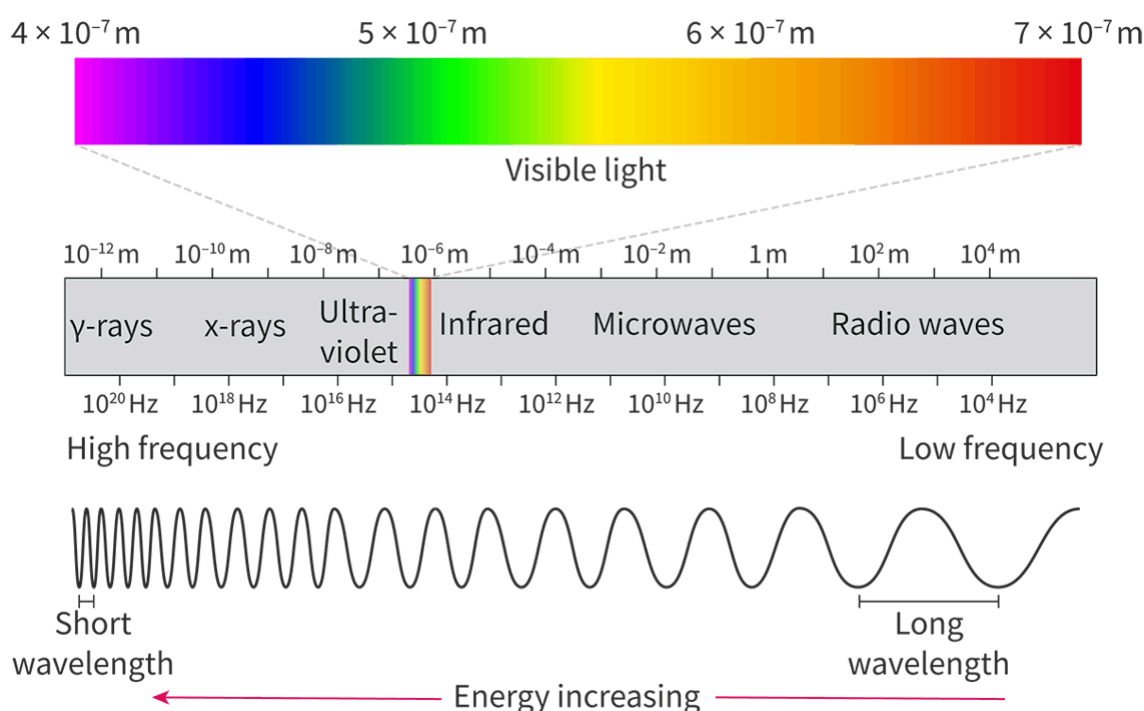
 More information for figure 1

The image shows two distinct wave patterns labeled Wave A and Wave B to illustrate the relationship between wavelength and frequency. Wave A is depicted with a longer wavelength and lower frequency, meaning the distance between consecutive crests is larger, and fewer waves pass a point per second. Wave B, on the other hand, has a

shorter wavelength and higher frequency, indicating a shorter distance between crests with more waves passing a point per second. Both waves are shown in a simple sinusoidal form on separate horizontal lines for clear comparison.

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The energy and frequency of electromagnetic radiation are directly proportional, meaning that if the energy increases so does the frequency. High energy corresponds to high frequencies and short wavelengths, and low energy corresponds to low frequencies and long wavelengths.



**Figure 2.** The electromagnetic spectrum.

 More information for figure 2

The image displays the electromagnetic spectrum with various types of waves represented, ranging from gamma rays on the left to radio waves on the right. At the top of the image, a gradient illustrates the change in wavelength and frequency, with short wavelengths and high frequencies on the left, transitioning to long wavelengths and low frequencies on the right. Beneath the gradient, different sections are labeled as gamma rays ( $\gamma$ -rays), X-rays, ultraviolet, visible light, infrared, microwaves, and radio waves. The visible light spectrum is highlighted with a small rainbow-colored band. Text at the bottom indicates areas of high energy/frequency and short wavelengths (on the left) and low energy/frequency and long wavelengths (on the right).

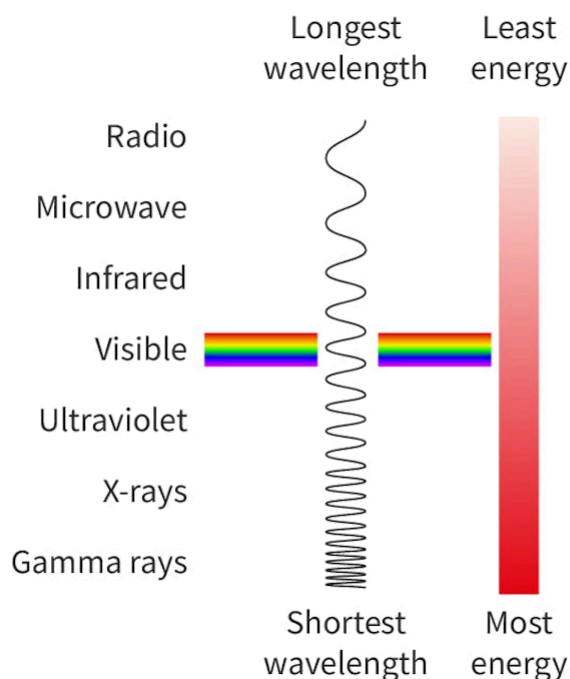
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## Study skills

The electromagnetic spectrum can be found in section 5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-electromagnetic-spectrum-id-45107/>) of the DP chemistry data booklet.

You should understand the terms directly proportional and inversely proportional. Make sure you are familiar with these terms and are able to use them correctly when explaining the relationship between frequency and wavelength, for example.

The seven regions of the electromagnetic spectrum are shown in **Figure 3**.



**Figure 3.** The seven regions of the electromagnetic spectrum.

 More information for figure 3

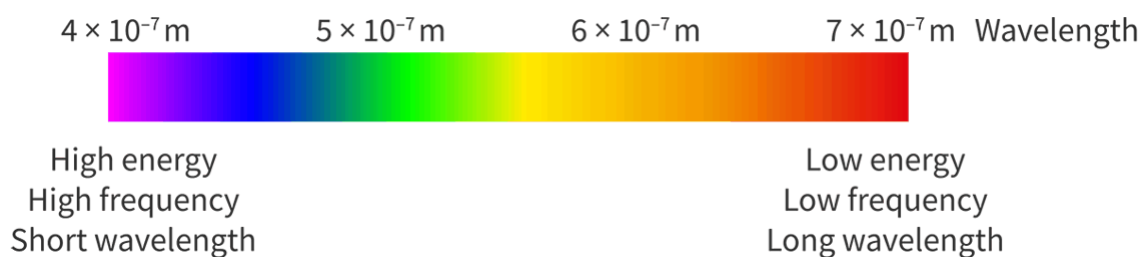
The diagram illustrates the seven regions of the electromagnetic spectrum, from radio waves to gamma rays. The spectrum is represented vertically with radio waves at the top and gamma rays at the bottom. To the left, labels indicate the types of radiation: Radio, Microwave, Infrared, Visible, Ultraviolet, X-rays, and Gamma rays. In the center, a sinusoidal waveform illustrates the varying wavelength, from longest at the top (radio) to shortest at the bottom (gamma rays). To the right, a gradient bar indicates energy levels, from least energy at the top to most energy at the bottom. Above the waveform,

the label 'Longest wavelength' appears, and below, 'Shortest wavelength.' To the right of the gradient bar, 'Least energy' is labeled at the top, transitioning to 'Most energy' at the bottom. The visible spectrum is highlighted in a rainbow of colors in the middle of the diagram. This visual representation shows how wavelength decreases and energy increases across the electromagnetic spectrum.

[Generated by AI]

Of the seven regions in the electromagnetic spectrum, radio waves have the lowest energy, lowest frequency and the longest wavelength. As we go across the spectrum, the energy and frequency both increase and the wavelength decreases. Therefore, gamma rays have the highest energy, highest frequency and the shortest wavelength.

**Figure 4** shows the visible region in more detail, which is known as the continuous spectrum. Our eyes see the continuous spectrum as white light, but as you can see it is actually made up of different colours.



**Figure 4.** The visible region of the electromagnetic spectrum.

 More information for figure 4

The image shows the visible region of the electromagnetic spectrum as a continuous band of colors transitioning smoothly from violet on the left to red on the right. The colors include violet, blue, green, yellow, orange, and red, representing the range of visible light that our eyes can perceive. The spectrum highlights the concept of white light as a mixture of these various colors, correlating with the description provided before and after the image.

From the text provided after the image, red light is noted to have the lowest energy, frequency, and longest wavelength, whereas violet light, on the opposite end of the spectrum, has the highest energy, frequency, and shortest wavelength. This detailed illustration emphasizes the gradual shift in energy levels and wavelengths across the visible spectrum, making it distinct and informative about the nature of light and color.

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In the visible region, red light has the lowest energy, lowest frequency and longest wavelength. As we go across the spectrum, the energy and frequency both increase and the wavelength decreases. So violet light has the highest energy, highest frequency and shortest wavelength.

In **Interactive 1**, drag and drop the seven regions of the spectrum, and the frequency and wavelength, into the correct places.

High energy  
frequency  
wavelength

Low energy  
frequency  
wavelength

|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|

Radio waves

UV

X-rays

Micro waves

$\gamma$  rays

IR

Low

High

Short

Long

R

G

B

Y

V

O

I

☒ Check

### Interactive 1. Electromagnetic spectrum drag and drop.

 More information for interactive 1

This interactive drag-and-drop exercise helps students understand the structure of the electromagnetic (EM) spectrum by engaging them in a drag-and-drop activity. The primary goal is to correctly place the six regions of the EM spectrum, as well as

corresponding labels for frequency, wavelength, and visible light colors, in the appropriate positions based on their relative energies.

At the very top of the interactive activity, you will see two labeled sides. On the left side, above the spectrum, is a label that reads “High energy,” and on the right side is a label that reads “Low energy.”

Directly beneath each of these energy labels are two blank spaces; these are for indicating the frequency and wavelength associated with that energy level. These blanks are meant to be filled in by the user during the drag-and-drop exercise using the provided labels: High, Low, Short, and Long. A zoomed-in view between the third and fourth regions is further divided into seven segments.

Users can drag and drop the six correct regions of the spectrum, arrange the seven segments of the zoomed-in visible light band, and identify whether the frequency is higher or lower and the wavelength is shorter or longer at both the high-energy and low-energy ends of the spectrum.

The draggable gray tiles include;

- Radio waves
- UV
- X-rays
- Microwaves
- $\lambda$  rays
- IR
- Low
- High
- Short
- Long

Below the main spectrum row is a second, smaller segmented strip with seven blank boxes. This section is for arranging the visible spectrum in the correct order from high to low energy.

Draggable color tiles provided include;

- R (Red)
- O (Orange)
- Y (Yellow)
- G (Green)
- B (Blue)
- I (Indigo)
- V (Violet)

This interactivity helps users understand the electromagnetic spectrum and recognize the properties of waves. At the bottom of the drag-and-drop interactive activity is a "Check" button, designed to allow learners to submit their completed arrangement of the electromagnetic spectrum and receive instant feedback.

Read below for the solution:

The electromagnetic spectrum is divided into six regions as follows:

$\lambda$  rays

X-rays

UV

IR

Microwaves

Radio waves

In high energy, the frequency is "High" and the wavelength is "Short."

In low energy, the frequency is "Low" and the wavelength is "Long."

The zoomed-in view between the third and fourth regions is divided into seven segments as follows:

V

I

B

G

Y

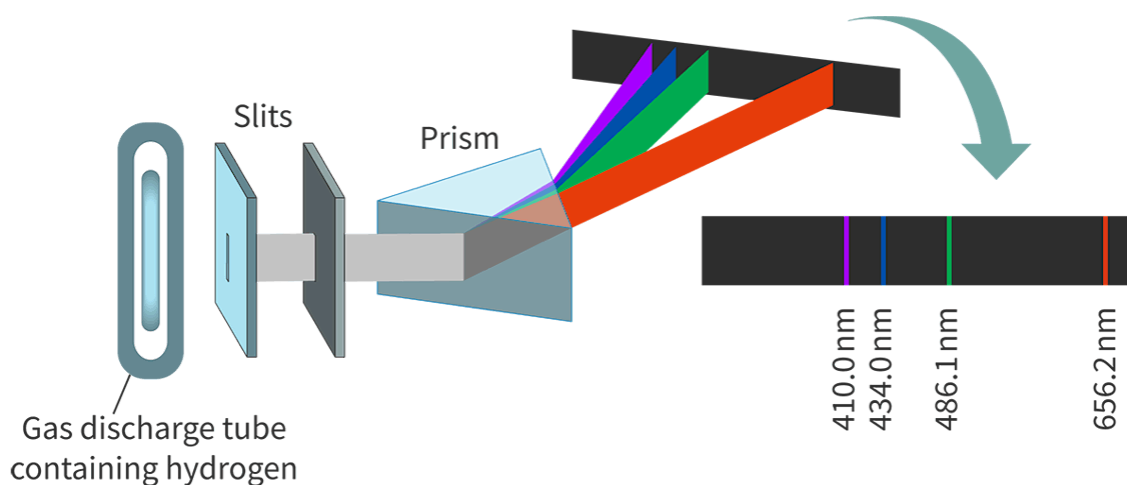
O

R

This activity provides a powerful visual framework for learners to understand the relationships among energy, frequency, and wavelength across the electromagnetic spectrum.

## Emission line spectra

When energy is applied to a sample of hydrogen gas, which is then viewed through a spectroscope, a distinctive pattern can be observed (**Figure 5**).



**Figure 5.** A gas discharge tube containing hydrogen produces a distinctive pattern.



 More information for figure 5

The image is a diagram illustrating the process inside a gas discharge tube containing hydrogen gas. On the left side, there is a tubular structure representing the gas chamber. As energy is applied, light is emitted, passing through various components that seem to include lenses and prisms or diffraction elements. This process results in the dispersion of light into a spectrum. The image shows distinct colored lines representing different wavelengths of light emitted by hydrogen. On the right side, vertical lines in colors such as red, blue, green, and violet represent the emission lines indicating specific wavelengths. These lines correspond to the unique emission line spectrum of hydrogen. The overall structure illustrates how light is collected, dispersed, and viewed, creating a spectrum similar to a fingerprint unique to hydrogen gas.

[Generated by AI]

The pattern observed is known as the emission line spectrum of hydrogen. The emission line spectrum produced is unique to hydrogen, just like fingerprints are to an individual. A comparison between an emission line spectrum and a continuous spectrum is shown in **Figure 6**. What are the differences that you can see between the two?



**Figure 6.** A comparison of a continuous spectrum (top) and an emission line spectrum (below).

 More information for figure 6

The image displays two types of spectra: a continuous spectrum at the top and an emission line spectrum below. The continuous spectrum shows a smooth gradient of colors ranging from violet to red, representing all visible wavelengths without interruption. In contrast, the emission line spectrum features distinct vertical lines at specific wavelengths against a dark background, corresponding to emission lines of hydrogen. These lines are seen at particular positions in blue, cyan, and red, reflecting the unique spectral fingerprint of hydrogen.

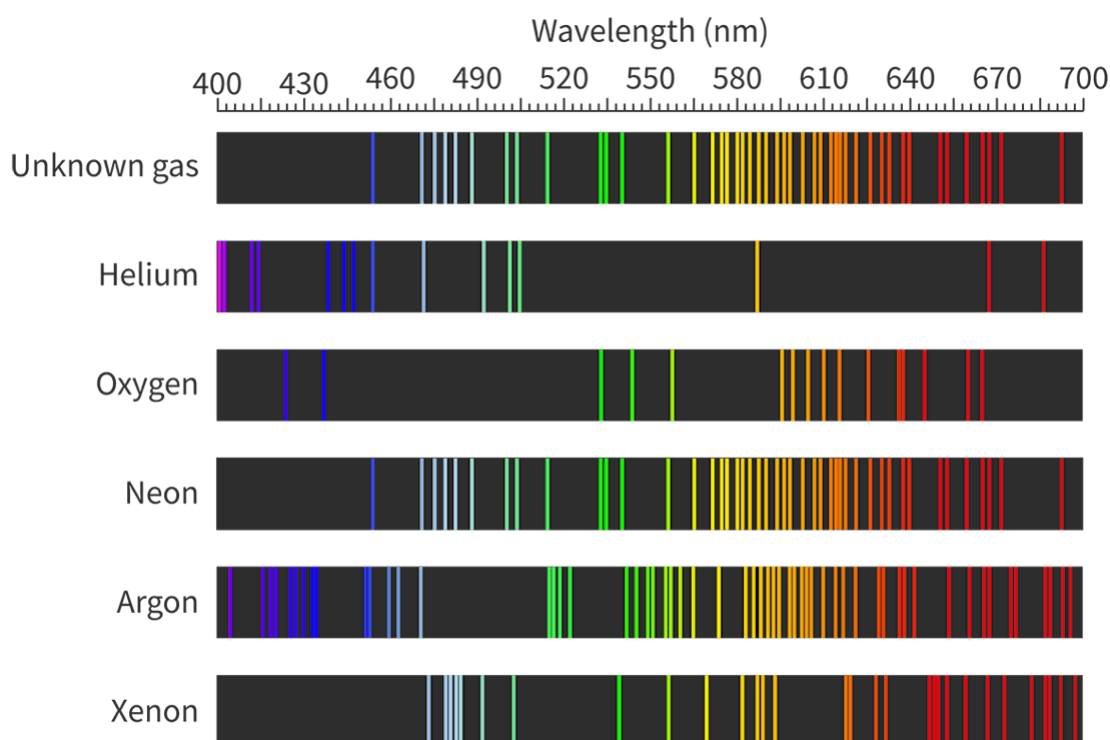
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- A continuous spectrum shows all the wavelengths or frequencies of visible light from red to violet.
- An emission line spectrum only shows specific wavelengths or frequencies of light. These are shown as coloured lines on a black background.

Looking at the emission line spectrum of hydrogen, what do you notice about spacing between the lines? Are they evenly spaced, or do they get closer together at one end of the spectrum? How does this observation correspond to what we have learned about the relationship between wavelength and energy in the visible light spectrum?

The lines in the emission spectrum get closer together (converge) at high energy, which corresponds to high frequency and short wavelength. So the distance between the blue and violet lines on the hydrogen emission spectrum is smaller than the distance between the red line and light blue line.

As mentioned before, each element produces its own unique emission line spectrum and these can be used to identify unknown elements. **Figure 7** shows a range of emission line spectra for a selection of elements together with an unknown element. From these can you identify the unknown element?



## Figure 7. A selection of emission line spectra for certain elements — can you identify the unknown gas?

 More information for figure 7

The image displays several emission line spectra, each consisting of colored lines arranged vertically on a black background. Each spectrum features lines in various positions indicating the spectral emissions of different elements. The colors range from purple and blue on the left to green, yellow, and red on the right. Each group of colored lines represents the unique emission spectrum of a specific element, aiding in the identification of an unknown gas by comparing its emission lines to these known examples.

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By comparing the emission line spectra, you can conclude that the unknown gas is neon.

### Practical skills

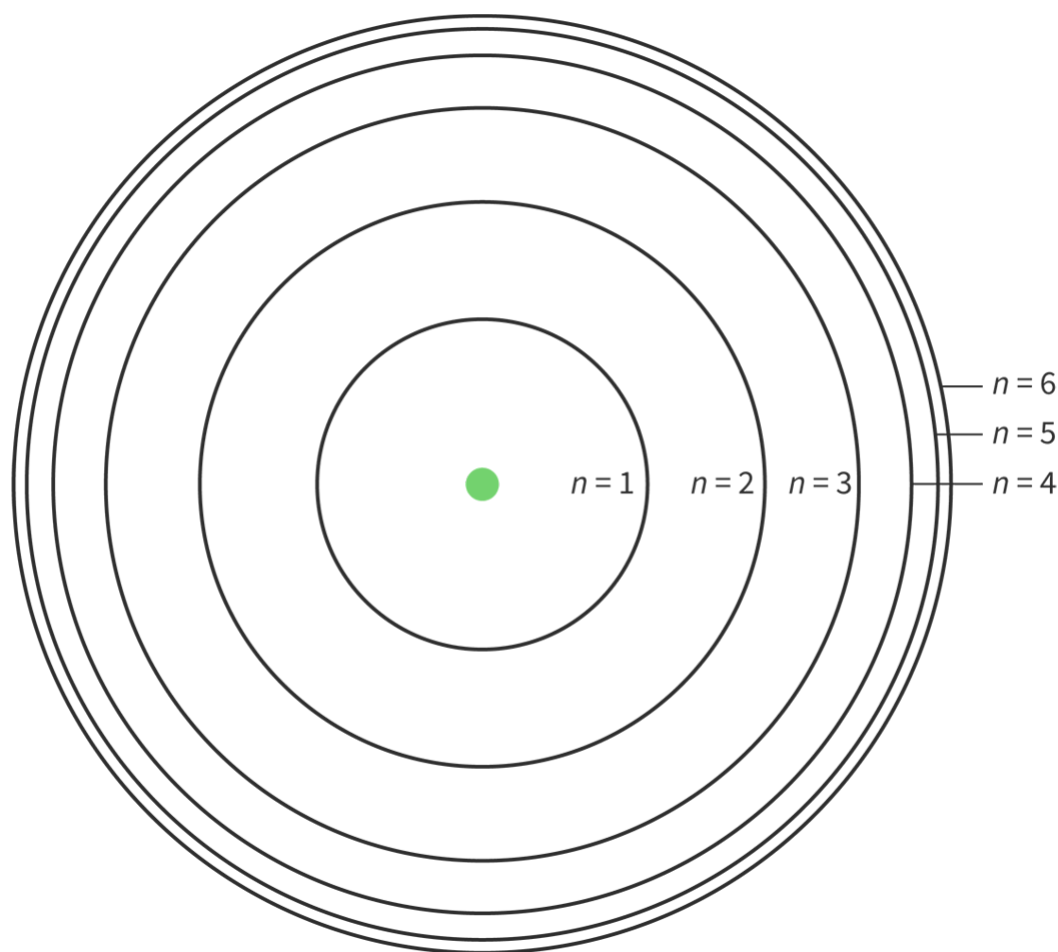
**Inquiry 2:** Collecting data — Identify and record relevant qualitative observations, Collect and record sufficient relevant quantitative data

In the study of emission spectra from gaseous elements and from light, what qualitative and quantitative data can be collected from instruments such as gas discharge tubes and prisms?

The formation of emission line spectra will be covered in more detail in section S1.3.3-4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>). For now, just consider which sub-atomic particle and which region of the electromagnetic spectrum is responsible for producing these line spectra?

To understand the formation of an emission spectrum requires us to revisit the model of the atom that was introduced in section S1.2.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/>), known as the Bohr model of the atom. Have you ever

wondered why electrons are not stuck to the nucleus? According to Bohr's model, electrons are located in energy levels surrounding the nucleus. **Figure 8** shows the energy levels of an atom (note that the contents of the nucleus are not shown as we are not concerned with them here).



**Figure 8.** The main energy levels in an atom from  $n = 1$  to  $n = 6$ .

 More information for figure 8

This diagram shows the main energy levels around an atom. It depicts six concentric circles around a central nucleus, representing energy levels from  $n=1$  to  $n=6$ . The first circle closest to the center is labeled  $n=1$ , indicating the ground state. As the circles extend outward, they are labeled consecutively with  $n=2$ ,  $n=3$ ,  $n=4$ ,  $n=5$ , and  $n=6$ , showing increasing energy levels further from the nucleus. The spacing between these circles decreases as they get further from the nucleus, illustrating the convergence of energy levels at higher levels, resembling the behavior observed in an emission spectrum where lines converge at higher energy levels. The diagram emphasizes the principle quantum number ' $n$ ' assigned to each energy level, indicating increased distances and energy states as the value of ' $n$ ' increases. This visualization aligns with the Bohr model's concept of electron energy levels in an atom.

[Generated by AI]

The energy levels are assigned the letter  $n$ , which is the principal quantum number. Energy level  $n = 1$  is closest to the nucleus (known as the ground state). As the value of  $n$  increases, so does the distance from the nucleus and its energy. So  $n = 1$  has the lowest energy and  $n = 6$  the highest. What do you notice about the spacing of the energy levels? Are they evenly spaced, or do they converge? Where have you seen this trend before? You may remember that the lines in an emission spectrum get closer together at high energy – this is because the main energy levels also converge at high energy.

### Making connections

You will learn more about how the energy levels are divided into sublevels in sections S1.3.3-4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>). Keep this model of the atom in mind or revisit this section to review when you get there.

Electrons can move (or transition) between the energy levels by either absorbing or emitting energy. The energy absorbed or emitted is in the form of photons, which are small packets of energy.

- If an electron absorbs a discrete or an exact amount of energy, it will transition from a lower energy level to a higher energy level, for example from  $n = 2$  to  $n = 3$ .
- The electron is now said to be in an excited state. The excited state is unstable relative to the ground state.
- The unstable electron emits the same amount of energy that it absorbed, and it transitions back down to  $n = 2$ .
- The amount of energy emitted by the electron in the transition from  $n = 3$  to  $n = 2$  corresponds to the wavelengths of visible light (in this example, red light).
- A red line will be observed on the emission line spectrum.

The amount of energy emitted also depends on the size of the transition. For example, more energy is emitted when an electron returns to  $n = 2$  from  $n = 5$  than is emitted when an electron returns to  $n = 2$  from  $n = 3$ . However, both of these transitions fall within the visible region as the energy they emit corresponds to the wavelengths of visible light. These electron transitions will be covered in more detail in the next section, S1.3.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/>).

**Interactive 2**, drag the slider across the screen to see how different transitions relate to different colours in the emission line spectrum.



### Interactive 2. Colours of the emission spectrum.

 More information for interactive 2

An interactivity allows users to explore how electronic transitions between energy levels correspond to different colors in the emission spectrum.

As the slider is dragged, different transitions appear, represented by downward arrows between energy levels labeled  $n = 6$  to  $n = 1$  (ground state). Each transition is visualized as colored lines in the spectrum at the bottom of the graph. The color of the lines and the transition of energy levels are as follows. Violet,  $n = 6$  to  $n = 2$ . Indigo,  $n = 5$  to  $n = 2$ . Blue,  $n = 4$  to  $n = 2$ . Red,  $n = 3$  to  $n = 2$ . Initially, no transitions are displayed, but as the slider is moved from right to left, the first visible transition from  $n = 3$  to  $n = 2$  appears, followed by additional transitions from higher energy levels to  $n = 2$ .

As the energy levels increase (from  $n = 1$  to  $n = 6$ ), the spacing between them decreases. This means that the energy difference between  $n = 4$  and  $n = 3$  is smaller than the difference between  $n = 3$  and  $n = 2$ . Consequently, transitions involving lower levels, such as from  $n = 3$  to  $n = 2$ , produce higher-energy photons, resulting in colors toward the blue and violet end of the spectrum. In contrast, transitions between higher levels (like  $n = 4$  to  $n = 3$ ) produce lower-energy photons, leading to colors toward the red end of the spectrum. This difference in energy differences explains why spectral lines from higher transitions appear farther to the left in the emission spectrum.

**Aspect: Evidence**

How do emission spectra provide evidence for the existence of different elements? As you have learned in this section, each element has a distinctive line spectrum just like each person has their own unique set of fingerprints. Throughout history, elements have been discovered from their emission line spectra; helium, discussed earlier, being one such example.

In this activity, you will consolidate your understanding about the relationship between energy, wavelength and frequency of electromagnetic radiation.

**Activity**

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Providing a reasoned argument to support conclusions
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual/pair activity

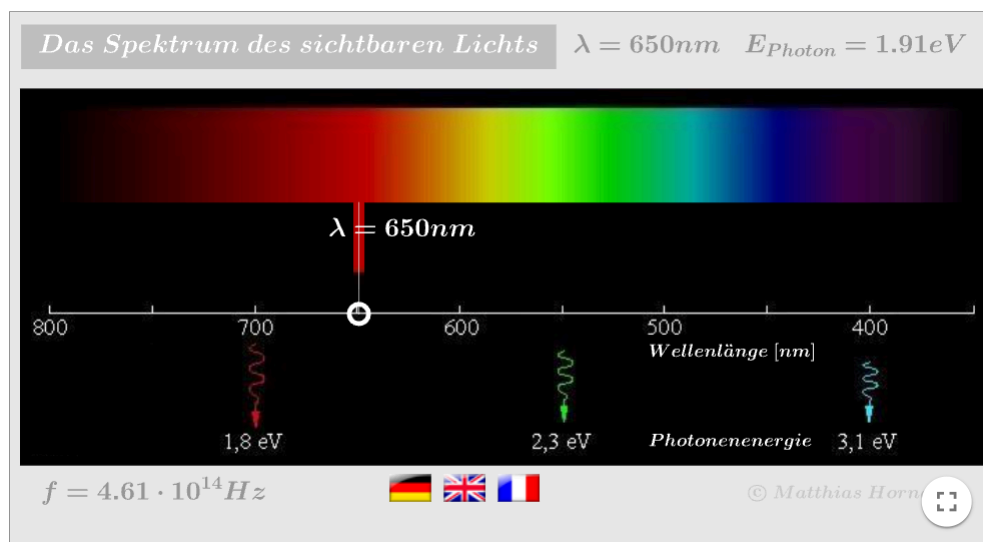
In this activity, you will consolidate your understanding about the relationship between energy, wavelength and frequency of electromagnetic radiation.

**Answer the following questions. Once you have finished, compare your answer with a partner.**

1. State the relationship between the frequency and wavelength of electromagnetic radiation.
2. State the relationship between the energy and wavelength of electromagnetic radiation.
3. State the relationship between the energy and frequency of electromagnetic radiation.
4. Decide whether the following statements are correct or incorrect. If incorrect, write the correct statement.
  - (a) Gamma rays have the highest energy, highest frequency and longest wavelength.
  - (b) Ultraviolet radiation has a shorter wavelength than X-rays.
  - (c) Microwaves have higher energy and a longer wavelength than infrared radiation.

- (d) Orange light is higher energy than green light.  
 (e) Blue light has a shorter wavelength and higher frequency than yellow light.

5. Outline two differences between an emission line spectrum and a continuous spectrum.  
 6. Examine the simulation below.



### Interactive 3. Energy, Wavelength, and Frequency of Electromagnetic Radiation.

🕒 More information for interactive 3

Move the slider all the way to the left (to the red light). Copy and complete the table below with the wavelength, frequency, energy of red light. Move the slider to the right and record the wavelength, frequency and energy of the remaining colours from orange to violet in the table.



| Colour of light | Wave (nm) |
|-----------------|-----------|
| Red             |           |
| Orange          |           |
| Yellow          |           |
| Green           |           |
| Blue            |           |
| Indigo          |           |
| Violet          |           |

7. From your table outline the relationship between the wavelength, frequency and energy of the visible region of the electromagnetic spectrum.

1. Frequency and wavelength are inversely proportional.
2. Energy and wavelength are inversely proportional.
3. Energy and frequency are directly proportional.
4. (a) Incorrect, gamma rays have the highest energy, highest frequency and shortest wavelength.  
 (b) Incorrect - ultraviolet radiation has a longer wavelength than X-rays.  
 (c) Incorrect - microwaves have a lower energy and longer wavelength than infrared radiation.

(d) Incorrect - orange light is lower energy than green light.

(e) Correct.

5. A continuous spectrum shows all the wavelengths or frequencies of visible light whereas an emission line spectrum has only certain wavelengths or frequencies (or colours) shown on a black background.
6. Answers will vary but you should see that as the frequency and energy increase, the wavelength decreases.
7. From red to violet, the energy and frequency both increase and the wavelength decreases.

# The hydrogen emission spectrum

## Learning outcomes

By the end of this section you should be able to describe the emission spectrum of the hydrogen atom, including the relationships between the lines and energy transitions to the first, second and third energy levels.

Hydrogen is the most abundant element – in fact it makes up approximately 90% of all the atoms in the universe. Hydrogen is located in group one of the periodic table with the alkali metals, even though it is not a metal. The reason for it being in group one is the fact that an atom of hydrogen only has one electron. In 1913, Niels Bohr proposed a model of the hydrogen atom, known as the planetary model.

- In this model the electrons were thought to travel around the nucleus in orbits (or shells), like planets orbiting the Sun.
- According to Bohr's model, only certain shells were allowed, and electrons could only exist within these shells and not between them.
- Electrons could move between the shells by either absorbing or emitting energy.

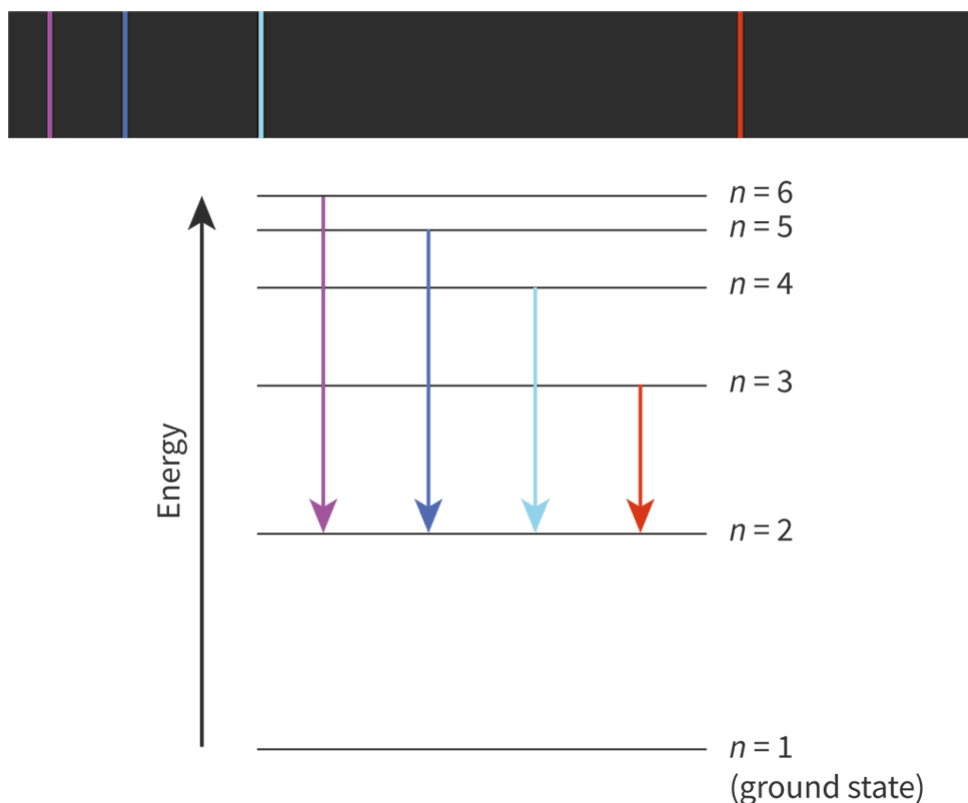
Using this model, Bohr was able to explain why hydrogen gas produced a distinctive emission line spectrum when a high voltage was passed through a sample of hydrogen gas at low pressure. However, Bohr's model only worked for atoms with one electron, namely hydrogen, but not for atoms with more than one electron.

## Making connections

In [section S1.3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>) you learned that the electrons are located in energy levels within the atom labelled  $n = 1$ ,  $n = 2$ , etc. If you need to review this section then now might be a good time to do so.

# Hydrogen emission spectrum

In the previous [section S1.3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>), you learned that emission line spectra are produced when electrons transition from higher to lower energy levels, emitting energy in the process. In this section we will focus on the emission spectra of one element in particular, hydrogen. The [hydrogen emission spectrum](#) in the visible region, together with the electron transitions responsible for each line, is shown in **Figure 1**.



**Figure 1.** The hydrogen emission spectrum in the visible region together with the electrons transition responsible for the lines.

 More information for figure 1

The image shows a diagram of the hydrogen emission spectrum in the visible region along with the electron transitions responsible for each spectral line. The spectrum is depicted with discrete colored lines representing different wavelengths corresponding to electron transitions from higher to lower energy levels in a hydrogen atom. These transitions are represented with arrows between energy levels. The lines' colors range from purple to red, indicating different energy transitions, such as the transition from  $n=3$  to  $n=2$ , which creates the red line in the spectrum, as explained in the surrounding text.

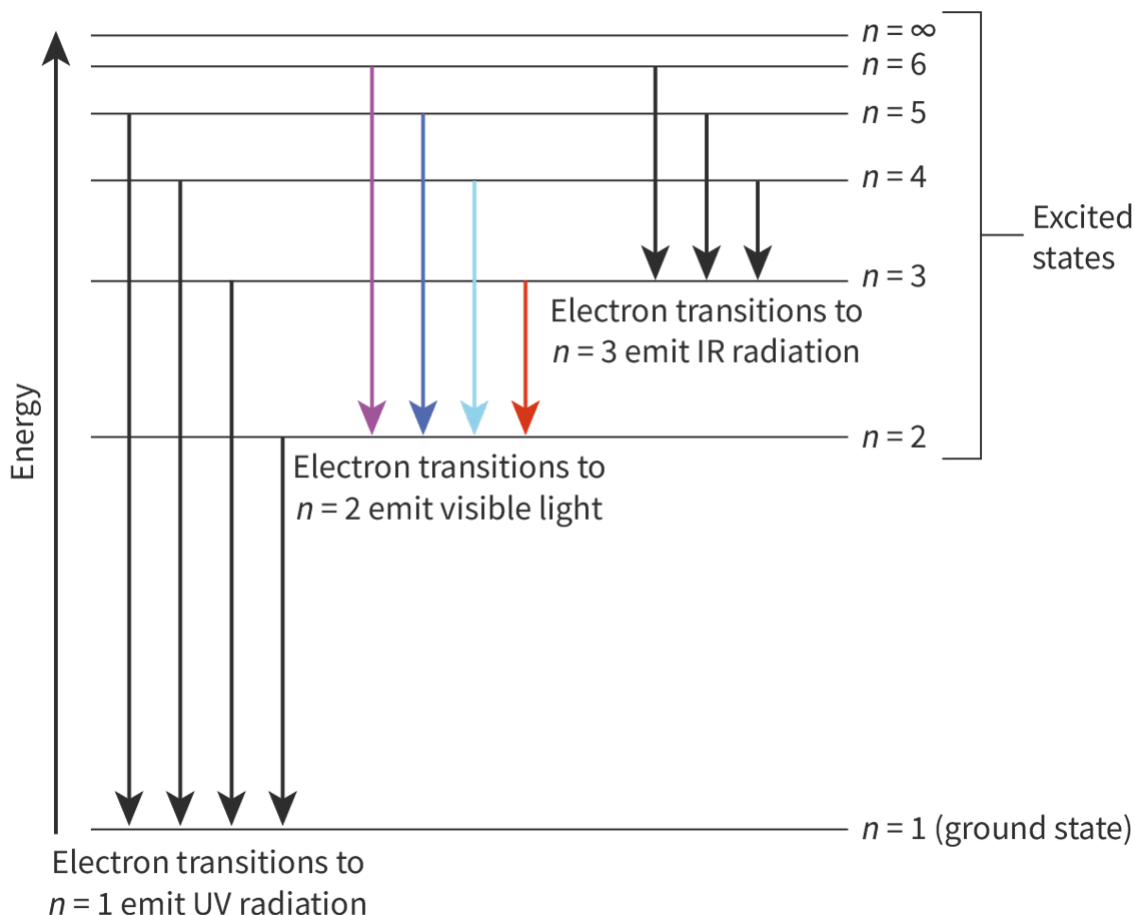
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Each electron transition corresponds to a specific line in the spectrum. For example, the electron transition from  $n = 3$  to  $n = 2$  produces the red line in the spectrum. Because we only see four discrete lines on the spectrum, this suggests that the electrons can only exist at certain energy levels. This provides evidence for the existence of energy levels in the atom. In other words, each line on the spectrum represents a discrete energy level in the hydrogen atom and electrons can only occupy these energy levels.

Finally, we will look at the complete hydrogen emission spectrum that shows the electron transitions from higher energy levels to the first, second and third main energy levels.

- Electron transitions from higher energy levels to  $n = 1$  emit energy that corresponds to UV radiation. These represent the highest energy transitions.
- Electron transitions from higher energy levels to  $n = 2$  emit energy that corresponds to visible light.
- Electron transitions from higher energy levels to  $n = 3$  emit energy that corresponds to infrared radiation. These represent the lowest energy transitions.

The full hydrogen emission spectrum is shown in **Figure 2**.



**Figure 2.** The hydrogen emission spectrum.

 More information for figure 2

The image is a diagram representing the hydrogen emission spectrum. It displays multiple vertical bands of color signifying different emission lines of hydrogen, such as the Lyman, Balmer, and Paschen series. Each band likely corresponds to specific wavelengths, visible as distinct color areas varied in length, representing the energy transitions within the hydrogen atom as electrons move between different energy levels. The spectrum may also suggest convergence at higher energy levels, indicating more closely spaced lines. The diagram illustrates these energy levels, possibly indicating a transition point where the electron is completely freed from the nucleus in ionization, labeled as  $n = \infty$ .

[Generated by AI]

A close look at the hydrogen emission spectrum reveals that the energy levels are not evenly spaced, instead they get closer together (converge) at high energies. There is also an energy level labelled as  $n = \infty$ . This represents the point when the electron has been completely removed from the attraction of the nucleus and the atom has been ionised.

## Making connections

The point at which the energy levels converge, the convergence limit, will be used in section S1.3.6 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/>) to calculate the ionisation energy (HL only).

In this activity, you will consolidate what you have learned in this section.

## Activity

- **IB learner profile attribute:** Thinkers
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Energy

Highest energy

Lowest energy

Highest frequency

Lowest frequency

Shortest wavelength

Longest wavelength

A

B

C

$n = 1$

$n = 2$

$n = 3$

$n = 4$

$n = 5$

$n = 6$

$n = \infty$

✓ Check

### Interactive 1. Label the Energy Levels.

More information for interactive 1

This interactive **energy level diagram** allows users to explore electron transitions in the hydrogen atom and understand their relationship with energy, frequency, and wavelength of emitted photons.

A vertical arrow on the left represents **increasing energy**. Horizontal lines represent discrete **energy levels** ( $n = 1$  to  $n = \infty$ ). The spacing between these lines **increases** as you move **down the diagram**, indicating that energy differences between lower levels are larger.

**Three downward arrows** show **electron transitions** from higher to lower energy levels.

**Users must drag and drop the correct labels to complete the diagram:**

1. Label all seven energy levels from bottom (ground state) to top (limit of ionization).
2. Label each of the three transitions with A, B, or C.
3. Order the transitions based on:



- Energy of emitted photons (highest to lowest)
- Frequency of emitted photons (highest to lowest)
- Wavelength of emitted photons (shortest to longest)

Drag-and-drop options provided are:

A

B

C

$n = 1$

$n = 2$

$n = 3$

$n = 4$

$n = 5$

$n = 6$

$n = \infty$

This interactivity helps users identify and label the energy levels and electronic transitions and understand the relationship between energy level transitions and the emitted photon's energy, frequency, and wavelength.

Read below for the solution:

#### Energy Levels (Bottom to Top):

- $n = 1$
- $n = 2$
- $n = 3$
- $n = 4$
- $n = 5$
- $n = 6$
- $n = \infty$

#### Electron Transitions:

- Arrow from  $n = 4 \rightarrow n = 3$  : C
- Arrow from  $n = 3 \rightarrow n = 2$  : B
- Arrow from  $n = 2 \rightarrow n = 1$  : A

#### Order of transitions:

Highest to lowest energy: A, B, C

Highest to lowest frequency: A, B, C

Shortest to longest wavelength: A, B, C

1. Label the energy levels on the diagram from  $n = 1$  to  $n = \infty$ .

2. Label each arrow with A, B or C.

A represents the electron transition that emits UV radiation,

B represents the electron transition that emits visible light.

C represents the electron transition that emits IR radiation.

3. Complete the table with the energy, frequency and wavelength of the three electron transitions A, B and C.

# Main energy levels and sublevels

## Learning outcomes

By the end of this section you should be able to:

- (S1.3.3) Deduce the maximum number of electrons that can occupy each energy level.
- (S1.3.4) Recognise the shape and orientation of an s atomic orbital and the three p atomic orbitals.

In [section S1.3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>) you learned that electrons are located in discrete energy levels called main energy levels, or principal energy levels. A rather interesting thing about electrons is that, because of their very small size and mass, we cannot know exactly where they are located. In 1927, a German physicist named [Werner Heisenberg](https://en.wikipedia.org/wiki/Werner_Heisenberg) ([https://en.wikipedia.org/wiki/Werner\\_Heisenberg](https://en.wikipedia.org/wiki/Werner_Heisenberg)) came up with the [Heisenberg uncertainty principle](#) which, in its simplest form, states that it is not possible to know both the location and [velocity](#) of very small particles such as electrons. So how do we overcome this? Rather than knowing exactly where an electron is, we assign regions of space, known as atomic orbitals, where we can be fairly certain that electrons are located. The Heisenberg uncertainty principle is outside of the scope of the IB chemistry syllabus and is not assessed, but if you are interested, you may read about it in more detail [here](#)  (<https://www.theguardian.com/science/2013/nov/10/what-is-heisenbergs-uncertainty-principle>).

## Making connections

In [section S1.3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>) you learned that the electrons are located in energy levels within the atom labelled  $n = 1$ ,  $n = 2$ , etc. If you need to review this section then now might be a good time to do so.

# Main energy levels

The main energy levels were introduced in [section S.1.3.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>). Recall that each energy level is assigned a number, starting with  $n = 1$  with the lowest energy and being closest to the nucleus. As the value of  $n$  increases, the energy increases and so does the distance from the nucleus. How do you think the electrons are organised in these energy levels? Do you think they are randomly arranged or is there a system of organisation?

To review the concept of main energy levels, try **Interactive 1** before you continue with this section. In the first part, drag the labels into the correct boxes. In the second part, fill in the blanks to complete the sentences.

Diagram illustrating the main energy levels of an atom, showing concentric circles representing energy levels around a central nucleus (green dot).

Labels to be dragged into the boxes:

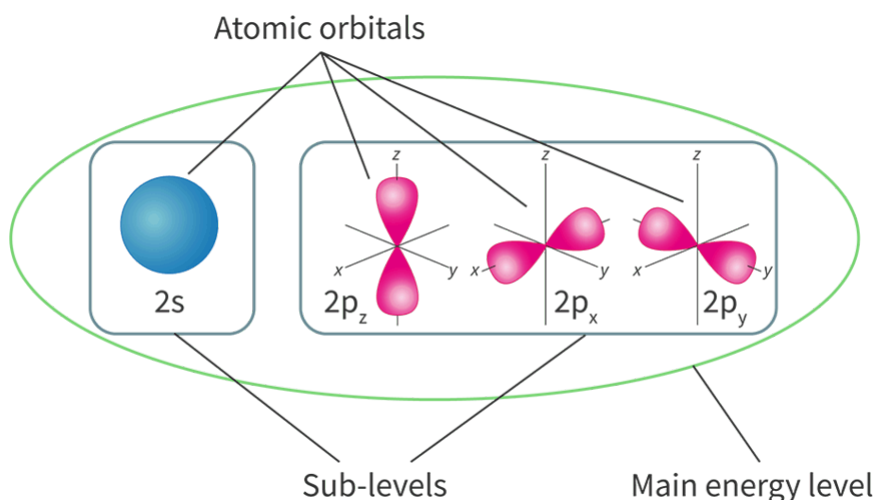
- $n = 1$
- $n = 2$
- $n = 3$
- $n = 4$
- $n = 5$
- Nucleus

Buttons: Check, >

## Interactive 1. Energy Levels Drag and Drop.

The main energy levels are divided into sublevels. The sublevels are assigned the letters s, p, d and f based on the shape of atomic orbitals they contain. Atomic orbitals can be thought of as regions of space where there is a high probability of finding an electron. A single atomic orbital can hold a maximum of two electrons and as you will learn in section S1.3.5

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/>), they must have specific orientations. So, to recap, we have main energy levels which are divided into sublevels which are made up of atomic orbitals (**Figure 1**).



**Figure 1.** The shapes of s and p atomic orbitals.

 More information for figure 1

The image shows a diagram illustrating the shapes of s and p atomic orbitals. On the left, the s orbital is represented as a blue sphere indicating its spherical shape. On the right, the p orbitals are depicted in pink with a dumbbell shape, demonstrating their characteristic form. The p orbitals are shown in three orientations, commonly labeled as  $p_x$ ,  $p_y$ , and  $p_z$ , reflecting their spatial orientation along the x, y, and z axes. This image explains the correlation between these orbital shapes and their electron-holding capacities: the s orbital, being singular, holds up to two electrons, whereas the three p orbitals collectively hold up to six electrons. The surrounding text provides additional context on energy levels and sublevels.

[Generated by AI]

**Figure 1** shows the shapes of two specific atomic orbitals: s atomic orbitals and p atomic orbitals. The s atomic orbitals are spherical and p atomic orbitals are dumbbell shaped. An s sublevel only consists of a single s atomic orbital so it can hold a maximum of two electrons. A p sublevel is composed of three p atomic orbitals so it can hold a maximum of six electrons.

There are also d and f atomic orbitals but the shapes of these are complex and are not covered in the IB chemistry course. A d sublevel contains five d atomic orbitals, and an f sublevel contains seven f atomic orbitals. Regardless of the type of atomic orbital (s, p, d or f) a single orbital can hold a maximum of two electrons. **Table 1** shows the first four main energy levels, together with the number of sublevels, atomic orbitals and total number of electrons.

**Table 1.** The first four main energy levels and orbitals

| Main energy level, $n$ | Sublevels      | Number of s orbitals | Number of p orbitals | Number of d orbitals | Number of f orbitals |
|------------------------|----------------|----------------------|----------------------|----------------------|----------------------|
| 1                      | 1s             | 1                    | 0                    | 0                    | 0                    |
| 2                      | 2s, 2p         | 1                    | 3                    | 0                    | 0                    |
| 3                      | 3s, 3p, 3d     | 1                    | 3                    | 5                    | 0                    |
| 4                      | 4s, 4p, 4d, 4f | 1                    | 3                    | 5                    | 7                    |

From the table, you can see that the maximum number of electrons in an energy level is  $2n^2$  where  $n$  is the principal energy level number. For the first main energy level,  $n = 1$ , the maximum number of electrons it can hold is  $2(1)^2$ , which is 2. For the second main energy level, it is  $2(2)^2$ , which is 8. The number of electrons increases as the value of  $n$  increases. So the  $n = 3$  energy level can hold a maximum of 18 electrons and the  $n = 4$  energy level a maximum of 32 electrons.

### Making connections

How does an element's highest main energy level relate to its period number in the periodic table? In [subtopic S3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/), you will learn about the relationship between the highest occupied main level of an element and its period number in the periodic table.

In this activity, you will determine the number of atomic orbitals in each of the main energy levels together with the shapes of the s and p atomic orbitals.

## Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Reflecting at all stages of the assessment and learning cycle
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Answer the following questions.

Check your answers once you have completed all of the questions.

1. Copy and complete the table with the maximum number of electrons in each main energy level.

| Principal energy level ( $n$ ) | Maximum number of electrons |
|--------------------------------|-----------------------------|
| 1                              |                             |
| 2                              |                             |
| 3                              |                             |
| 4                              |                             |

2. Draw a table like the one below.

Sketch the shape of an s and p atomic orbital in the table.

| s orbital |  |
|-----------|--|
|           |  |

|                  |  |
|------------------|--|
| <b>s orbital</b> |  |
|                  |  |

3. State the number of orbitals in each of the following sublevels.

- (a) The 2s sublevel
- (b) The 3p sublevel
- (c) The 3d sublevel
- (d) The 4f sublevel

4. State the total number of each type of atomic orbital (s, p, d, f) in each main energy level.

- (a)  $n = 1$
- (b)  $n = 2$
- (c)  $n = 3$
- (d)  $n = 4$

1.

| Principal energy level ( $n$ ) | M |
|--------------------------------|---|
| 1                              |   |
| 2                              |   |
| 3                              |   |
| 4                              |   |

2.

| s orbital                                                                           | p orbital                                                                            |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |  |





3. (a) The 2s sublevel (one s orbital)  
(b) The 3p sublevel (three p orbitals)  
(c) The 3d sublevel (five d orbitals)  
(d) The 4f sublevel (seven f orbitals)
4. (a)  $n = 1$  (one s orbital)  
(b)  $n = 2$  (one s orbital and three p orbitals)  
(c)  $n = 3$  (one s orbital, three p orbitals and five d orbitals)  
(d)  $n = 4$  (one s orbital, three p orbitals, five d orbitals and seven f orbitals)

# Electron configurations

## Learning outcomes

By the end of this section you should be able to apply the Aufbau principle, Hund's rule and the Pauli exclusion principle to deduce electron configurations for atoms and ions up to  $Z = 36$ .

When sending a letter or parcel, it is necessary to write the address to where it is to be delivered. An address usually has the postal code, city name, as well as the street name and house name or number. Although we can't know exactly where electrons are located in an atom, we can write an electron configuration for an atom. This is similar to an address and tells us where the electrons are located – in which main energy level and sublevel – as well as how many electrons there are in that particular region. In this section, you will learn how to write and interpret electron configurations for atoms and ions.

## Electron configurations

An atomic orbital is a region of space inside the atom where there is a high probability of finding an electron. In section S1.3.3–4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>) you learned the shapes and s and p atomic orbitals. Recall that there are also d and f orbitals but the shapes of these are not required.

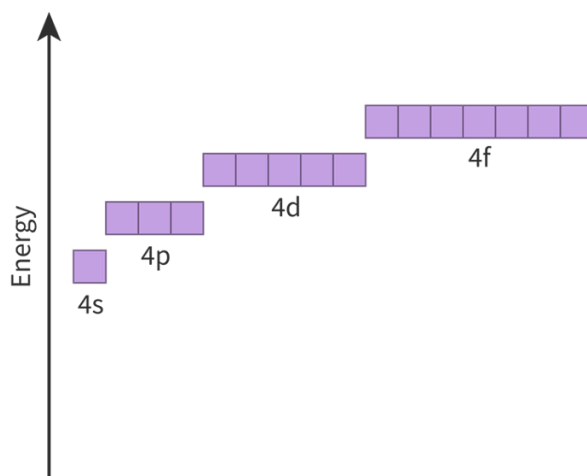
To fill atomic orbitals with electrons, we must follow certain principles, which are summarised here.

- The Aufbau principle states that when adding electrons to an atom, the lower energy orbitals must be filled first.
- The Pauli exclusion principle states that an atomic orbital can only hold two electrons and they must have opposite spins.
- Hund's rule states that when we have degenerate orbitals (orbitals of the same energy) then each orbital is filled with a single electron before being doubly occupied.

For example, a helium atom contains two electrons in the first energy level in an s orbital – this electron configuration would be stated as  $1s^2$ . The '1' represents the energy level, 's' the shape of the orbital and '2' the number of electrons. You will learn how to state the electron configurations for atoms with more electrons below.

# Energy level diagrams

Within a given main energy level, s orbitals are of lower energy than p orbitals which are of lower energy than d orbitals. f orbitals have the highest energy. So the order of energy of the sublevels is  $s < p < d < f$ . **Figure 1** shows the energy of the 4s, 4p, 4d and 4f orbitals in the  $n = 4$  energy level.



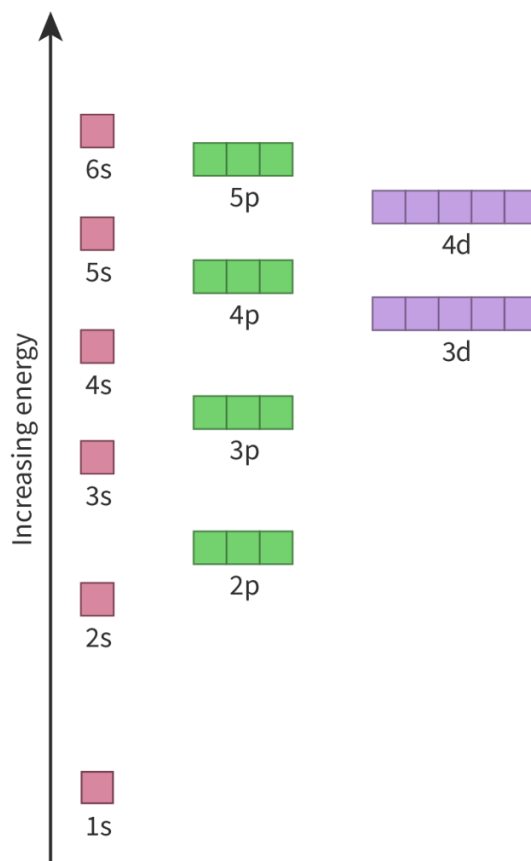
**Figure 1.** The 4s, 4p, 4d and 4f atomic orbitals in the  $n = 4$  energy level.

 More information for figure 1

The image is a diagram illustrating the energy levels of atomic orbitals in the  $n=4$  energy level. It shows a stepped arrangement, with each step representing an orbital sublevel: 4s, 4p, 4d, and 4f. The 4s orbital is depicted at the lowest energy level, followed by 4p, 4d, and 4f, which is at the highest level. Each orbital is represented by rectangles aligned on horizontal planes, indicating the relative energy and inclusion of electron occupancy levels.

[Generated by AI]

The order of the sublevels from 1s to 6s in terms of their energy is shown in **Figure 2**.



**Figure 2.** The 1s to 6s atomic orbitals in order of increasing energy.

 More information for figure 2

The image is a diagram depicting the order of atomic orbitals from 1s to 6s in terms of increasing energy. On the left, there is a vertical arrow labeled 'Increasing energy'. Next to the arrow, boxes representing the s, p, and d orbitals are arranged in columns. The 1s orbital is at the bottom, followed by 2s, 2p, 3s, 3p, 3d, 4s, 4p, 4d, 5s, 5p, and finally 6s at the top. Each of these orbitals is illustrated by a series of small boxes, where s orbitals have a single box, p orbitals have three boxes, and d orbitals have five boxes, suggesting the number of electrons each can hold. The diagram visually represents the relative energy levels of these atomic orbitals as they ascend the chart.

[Generated by AI]

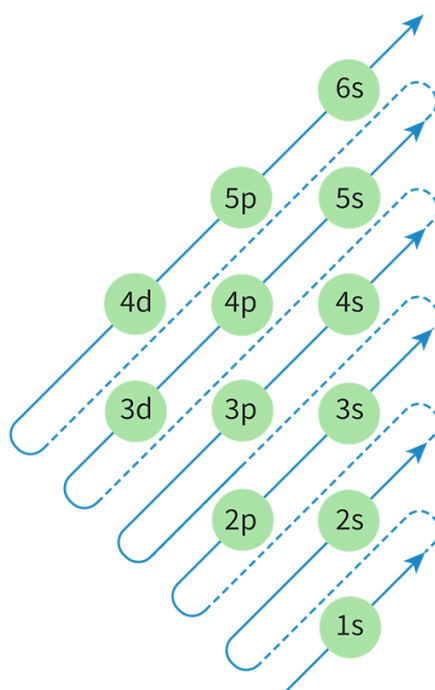
Here are some important points to note about the energy of the sublevels.

- The atomic orbitals within the same sublevel are of equal energy, known as degenerate orbitals. For example, the three p orbitals in the 2p sublevel are of equal energy. This also applies to the three p orbitals in the 3p sublevel and the five d orbitals in the 3d sublevel, etc.

- There is an overlap in energy between the 3d and 4s sublevels. The 4s sublevel is of lower energy and fills before the 3d sublevel\*. This has important consequences when writing the electron configuration of the d-block elements.

\*The above paragraph states that the 4s sublevel fills before the 3d sublevel, but this is not the case for all atoms. The reasons for this are beyond SL and HL chemistry so will not be covered in this text. Students should be aware that the 4s sublevel fills before the 3d sublevel and electrons are removed from the 4s sublevel first when forming ions. If you are interested in reading more about this phenomenon, you can do so [here](https://ericscerri.blogspot.com/2012/06/trouble-with-using-aufbau-to-find.html) (<https://ericscerri.blogspot.com/2012/06/trouble-with-using-aufbau-to-find.html>).

The order of filling of the sublevels is summarised in **Figure 3**. This can be learned to help you remember the order of filling. Alternatively, another method can be used, known as the block method. This is covered in [subtopic S3.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>).



**Figure 3.** The order of filling of the sublevels 1s through to 6s.

 More information for figure 3

The diagram illustrates the order of filling electron sublevels in an atom. The layout includes sublevels labeled as 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, and 6s. Each sublevel is represented in a separate section with a label, and arrows indicate the sequence in which electrons fill these sublevels, starting from 1s in the bottom right to 6s at the top right.

[Generated by AI]

Test your understanding by trying **Interactive 1**. Drag and drop the components in the correct order to make the electron configuration of selenium, Se.

2p<sup>6</sup>

4s<sup>2</sup>

1s<sup>2</sup>

3s<sup>2</sup>

2s<sup>2</sup>

3d<sup>10</sup>

3p<sup>6</sup>

4p<sup>4</sup>

|

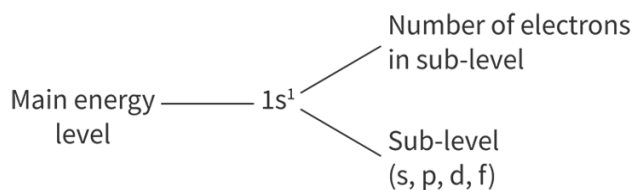
☐ Check

### Interactive 1. Electron Configuration of Selenium, Se.

#### Making connections

In section S3.1.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrons-and-the-periodic-table-id-44056/>) you will learn an alternative method to determine the electron configuration of an element using the periodic table. You can use whichever method you find easier as both give the correct electron configurations.

**Figure 4** shows the notation used when writing electron configurations. The number in front of the letter is the principal quantum number,  $n$ , which gives the number of the main energy level. The letter refers to the sublevel (s, p, d, f) and the number in superscript gives the number of electrons in the sublevel.



**Figure 4.** The notation used for electron configurations.

More information for figure 4

This is a diagram illustrating the notation used for writing electron configurations. It outlines the principal quantum numbers represented by numbers, which denote the main energy levels: 1, 2, 3, 4, etc. The diagram shows the sublevels as letters: s, p, d, f,

corresponding to different shapes or types of electron orbitals. Accompanying these letters, there are superscript numbers that denote the number of electrons within each sublevel. The flow of the diagram progresses from the lowest energy level to the highest, demonstrating a systematic buildup of electrons according to quantum mechanics' principles. Key segments of the diagram include the layout of energy levels vertically, the breakdown of each level into sublevels, and the electron count in superscript. This visual representation helps in understanding the electronic configuration of elements.

[Generated by AI]

## Worked example 1

Write the full electron configurations for the first ten elements of the periodic table, from hydrogen to neon.

Starting with hydrogen, H, which is atomic number 1. Atoms have the same number of protons and electrons, so hydrogen has 1 electron.

According to the Aufbau principle we add electrons to the lowest energy orbitals first, so we put one electron in the 1s orbital. The electron configuration of hydrogen is:



Next is helium, He, with the atomic number 2 which means helium has 2 electrons. This gives the electron configuration:



Next is lithium, Li, with atomic number 3 meaning it has 3 protons and 3 electrons. The 1s orbital can only hold two electrons so we need to start filling the 2s sublevel. The electron configuration of lithium is:

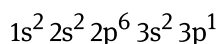


From here on, we will just add one to the atomic number and one to the number of electrons. We will also follow the Aufbau principle of filling lowest energy orbitals first.

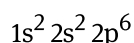
| Element       | Atomic number | Ele |
|---------------|---------------|-----|
| Beryllium, Be | 4             |     |
| Boron, B      | 5             |     |

| Element     | Atomic number | Ele |
|-------------|---------------|-----|
| Carbon, C   | 6             |     |
| Nitrogen, N | 7             |     |
| Oxygen, O   | 8             |     |
| Fluorine, F | 9             |     |
| Neon, Ne    | 10            |     |

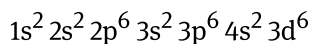
When writing the electron configurations of ions, remember to either add the correct number of electrons (according to the charge) for negative ions or remove the correct number of electrons for positive ions. Note that electrons are always removed from higher energy orbitals first. For example, aluminium, which forms a 3+ ion,  $\text{Al}^{3+}$ , has the electron configuration:



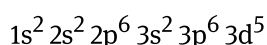
From which orbital would the first electron be removed? The first electron is removed from the 3p orbital because it is the highest energy orbital. The next two electrons are removed from the 3s orbital to give a total of three electrons removed. Therefore, the electron configuration of the  $\text{Al}^{3+}$  ion is:



For the transition elements, it is important to remember that electrons are removed from the 4s orbital first, and then from the 3d sublevel. For example, iron, Fe has the electron configuration:



The iron atom can lose three electrons to form the  $\text{Fe}^{3+}$  ion. Because the 4s orbitals have higher energy than the 3d orbitals, the two 4s electrons are removed first. The next electron is lost from the 3d sublevel, so the electron configuration of the  $\text{Fe}^{3+}$  ion is:



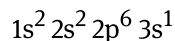
## Worked example 2

Write the full electron configurations for the following ions.

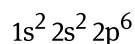


- (a)  $\text{Na}^+$
- (b)  $\text{Mg}^{2+}$
- (c)  $\text{F}^-$
- (d)  $\text{O}^{2-}$
  
- (e)  $\text{Ni}^{2+}$

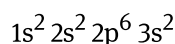
(a) Start by writing the electron configuration for the sodium atom.



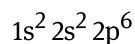
$\text{Na}^+$  is a cation with a 1+ charge meaning that it has one less electron than protons. We remove outer electrons first so the electron configuration of  $\text{Na}^+$  is:



(b) Following the same method for  $\text{Mg}^{2+}$ , we start with the full electron configuration of the magnesium atom:

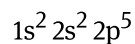


Because it is a cation with a 2+ charge, we then remove the two outer electrons in the 3s sublevel. The electron configuration of  $\text{Mg}^{2+}$  is:

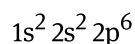


Note that the electron configurations for  $\text{Na}^+$  and  $\text{Mg}^{2+}$  are the same; known as being isoelectronic (have the same electron configuration).

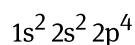
(c)  $\text{F}^-$  is an anion so we need to add electrons to the electron configuration of the fluorine atom, which is:



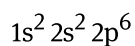
The  $\text{F}^-$  ion has a 1- charge so we add one electron to the 2p sublevel:



(d)  $\text{O}^{2-}$  is an anion so we need to add electrons to the electron configuration of the oxygen atom, which is:

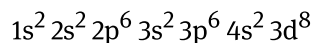


The  $O^{2-}$  ion has a 2- charge so we add two electrons to the 2p sublevel:

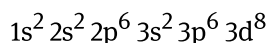


Once again you will notice that all of the ions in the example are also isoelectronic.

(e) The electrons configuration of the nickel atom is:

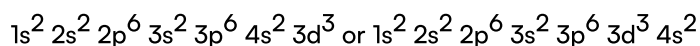


The  $Ni^{2+}$  ion has a 2+ charge so two electrons are removed from the 4s sublevel to give the electron configuration:



### Study skills

Although the 4s sublevel fills before the 3d sublevel, the electron configuration can be written with the 4s before, or after, the 3d sublevel. For example, the electron configuration of vanadium, V, can be written as:



Both ways of writing are accepted as being correct.

For atoms or ions with higher atomic numbers, we can write the condensed electron configurations. This notation uses the symbol of a noble gas to represent the core electrons. **Table 1** shows the noble gas symbol and the core electrons represented by that symbol.

**Table 1.** Noble gas symbols and the core electrons represented by the symbols.

| Symbol | Core electrons             |
|--------|----------------------------|
| [He]   | $1s^2$                     |
| [Ne]   | $1s^2 2s^2 2p^6$           |
| [Ar]   | $1s^2 2s^2 2p^6 3s^2 3p^6$ |

### Worked example 3

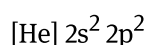
Write the condensed electron configurations for the following atoms.

- (a) Carbon, C
- (b) Phosphorus, P
- (c) Potassium, K
- (d) Cobalt, Co
- (e) Bromine, Br

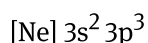
For each example, we will start by writing the full electron configuration and then write the condensed electron configuration using the noble gas symbol to represent the core electrons.

(a) For carbon, C, the full electron configuration is  $1s^2 2s^2 2p^2$

The core electrons are the electrons in the 1s sublevel so the noble gas symbol is [He]. The condensed electron configuration is:



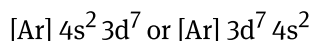
(b) For phosphorus, P, the full electron configuration is  $1s^2 2s^2 2p^6 3s^2 3p^3$ . The core electrons are the electrons in the 1s, 2s, and 2p sublevels so the noble gas symbol is [Ne]. The condensed electron configuration is:



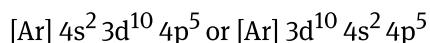
(c) For potassium, K, the full electron configuration is  $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ . The core electrons are the electrons in the 1s, 2s, 2p, 3s and 3p sublevels so the noble gas symbol is [Ar]. The condensed electron configuration is:



(d) For cobalt, Co, the full electron configuration is  $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$ . The core electrons are the electrons in the 1s, 2s, 2p, 3s and 3p sublevels so the noble gas symbol is [Ar]. The condensed electron configuration is:

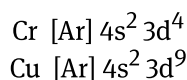


(e) For bromine, Br, the full electron configuration is  $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$ . The core electrons are the electrons in the 1s, 2s, 2p, 3s and 3p sublevels so the noble gas symbol is [Ar]. The condensed electron configuration is:



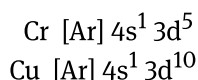
# Exceptions to the Aufbau principle

There are two important exceptions to the Aufbau principle that you should be aware of. These are for the elements copper (symbol Cu) and chromium (symbol Cr). Following the order of filling according to the Aufbau principle you would *expect* the electron configurations to be as follows:



However, the observed configurations are different from what is expected.

The actual electron configurations for copper and chromium are:



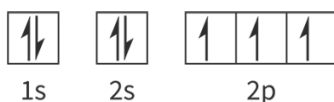
For chromium, there is one electron in the 4s sublevel and five in the 3d sublevel. And for copper there is one electron in the 4s sublevel and ten electrons in the 3d sublevel. You should take care to remember these exceptions, but no explanation is required.

## Theory of Knowledge

Rules and patterns, such as Hund's rule, the Aufbau principle and the Pauli exclusion principle are used to fill atomic orbitals with electrons for elements. However, rules often have exceptions, as you learned above with the examples of chromium and copper. Several other exceptions for electron configuration also occur in elements containing electrons in the d and f sublevels. Do exceptions make a rule less valid?

# Orbital diagrams

Electrons in atomic orbitals can be represented by orbital diagrams, also known as arrow in-box diagrams. These diagrams use boxes to represent atomic orbitals and single-headed arrows to represent electrons. According to the Pauli exclusion principle we can only put two arrows in a box, and they must have opposite spins – this is represented by one arrow pointing up and one arrow pointing down. For example, the electron configuration of nitrogen, N, is  $1s^2 2s^2 2p^3$ . The arrow in box diagram for nitrogen is shown in **Figure 5**. Note that it is important how we place the arrows (electrons) in the boxes (orbitals) as will be discussed next.



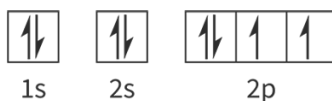
**Figure 5.** The orbital diagram for nitrogen, N.

 More information for figure 5

The image represents an orbital diagram for nitrogen (N). It consists of a series of boxes and arrows, indicating the arrangement of electrons within nitrogen's atomic orbitals according to the arrow-in-box format. The configuration begins with the 1s orbital, which contains two arrows indicating two electrons with opposite spins, one pointing upwards and one downwards. This is followed by the 2s orbital, also containing two electrons with opposite spins. Following are three 2p orbitals, each with a single arrow pointing upwards, representing nitrogen's three unpaired electrons according to Hund's rule.

[Generated by AI]

In the 2p sublevel you can see that we put one electron in each 2p atomic orbital according to Hund's rule. This states that degenerate orbitals are singly occupied before being doubly occupied. The next element, oxygen, has the electron configuration  $1s^2 2s^2 2p^4$ . The orbital diagram of oxygen, O, is shown in **Figure 6**. From this you can see that once each orbital has a single electron in it, we then start doubly occupying each orbital.



**Figure 6.** The orbital diagram for oxygen, O.

 More information for figure 6

The image is an orbital diagram for oxygen representing its electron configuration. It shows that the oxygen atom has three main sections for sublevels: 1s, 2s, and 2p. The 1s and 2s sublevels each have a pair of dots indicating that they are fully occupied with two electrons. The 2p sublevel has three parts, each represented by a box. One electron occupies each of the three boxes, following Hund's rule, and then one of these boxes has a pair of electrons, illustrating the electron configuration  $1s^2 2s^2 2p^4$ . This arrangement shows that the last electron is added to one of the already singly-occupied 2p orbitals, resulting in one full box in the 2p sublevel and two half-full boxes, matching the electron count of an oxygen atom.

[Generated by AI]

## Making connections

What is the relationship between energy sublevels and the block nature of the periodic table? In [subtopic S3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/) you will learn about the relationship between the number of occupied energy levels in an atom and its position in the periodic table.

In this activity you will practise writing electron configurations.

## Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Reflecting at all stages of the assessment and learning cycle
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

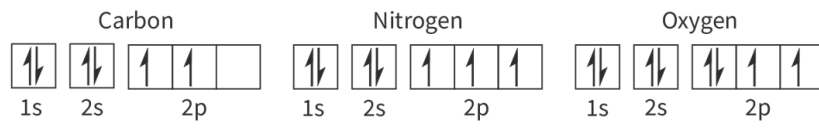
In this activity, you will practise writing electron configurations for atoms. Write full and condensed electron configurations for the following atoms.

| Atoms  |        |
|--------|--------|
| 1. He  | 11. Ar |
| 2. Li  | 12. Ca |
| 3. B   | 13. Ti |
| 4. C   | 14. Mn |
| 5. O   | 15. Ni |
| 6. Ne  | 16. Zn |
| 7. Na  | 17. Ge |
| 8. Al  | 18. Se |
| 9. P   | 19. Br |
| 10. Cl | 20. Kr |

Draw orbital diagrams for the following atoms.

- (a) Carbon
- (b) Nitrogen
- (c) Oxygen

| Electron Configuration                                              |                                                                        |
|---------------------------------------------------------------------|------------------------------------------------------------------------|
| 1. He: $1s^2$                                                       | 11. Ar: $1s^2 2s^2 2p^6 3s^2$<br>or<br>$[\text{Ne}] 3s^2 3p^6$         |
| 2. Li: $1s^2 2s^1$<br>or<br>$[\text{He}] 2s^1$                      | 12. Ca: $1s^2 2s^2 2p^6 3s^2$<br>or<br>$[\text{Ar}] 4s^2$              |
| 3. B: $1s^2 2s^2 2p^1$<br>or<br>$[\text{He}] 2s^2 2p^1$             | 13. Ti: $1s^2 2s^2 2p^6 3s^2$<br>or<br>$[\text{Ar}] 4s^2 3d^2$         |
| 4. C: $1s^2 2s^2 2p^2$<br>or<br>$[\text{He}] 2s^2 2p^2$             | 14. Mn: $1s^2 2s^2 2p^6 3s^2$<br>or<br>$[\text{Ar}] 4s^2 3d^5$         |
| 5. O: $1s^2 2s^2 2p^4$<br>or<br>$[\text{He}] 2s^2 2p^4$             | 15. Ni: $1s^2 2s^2 2p^6 3s^2$<br>or<br>$[\text{Ar}] 4s^2 3d^8$         |
| 6. Ne: $1s^2 2s^2 2p^6$<br>or<br>$[\text{He}] 2s^2 2p^6$            | 16. Zn: $1s^2 2s^2 2p^6 3s^2$<br>or<br>$[\text{Ar}] 4s^2 3d^{10}$      |
| 7. Na: $1s^2 2s^2 2p^6 3s^1$<br>or<br>$[\text{Ne}] 3s^1$            | 17. Ge: $1s^2 2s^2 2p^6 3s^2$<br>or<br>$[\text{Ar}] 4s^2 3d^{10} 4p^2$ |
| 8. Al: $1s^2 2s^2 2p^6 3s^2 3p^1$<br>or<br>$[\text{Ne}] 3s^2 3p^1$  | 18. Se: $1s^2 2s^2 2p^6 3s^2$<br>or<br>$[\text{Ar}] 4s^2 3d^{10} 4p^4$ |
| 9. P: $1s^2 2s^2 2p^6 3s^2 3p^3$<br>or<br>$[\text{Ne}] 3s^2 3p^3$   | 19. Br: $1s^2 2s^2 2p^6 3s^2$<br>or<br>$[\text{Ar}] 4s^2 3d^{10} 4p^5$ |
| 10. Cl: $1s^2 2s^2 2p^6 3s^2 3p^5$<br>or<br>$[\text{Ne}] 3s^2 3p^5$ | 20. Kr: $1s^2 2s^2 2p^6 3s^2$<br>or<br>$[\text{Ar}] 4s^2 3d^{10} 4p^6$ |





# Calculating ionisation energies (HL)

## Higher level (HL)

### Learning outcomes

By the end of this section you should be able to:

- Explain the trends and discontinuities in first ionisation energy (IE) across a period and down a group.
- Calculate the value of the first IE from spectral data that gives the wavelength or frequency of the convergence limit.

What type of forces hold an atom together? So far you have learned that an atom contains three sub-atomic particles – the proton, neutron and electron. The protons and neutrons are located in the nucleus and are held together by the strong nuclear force. This strong force is required to overcome the repulsion between the positively charged protons – without it the nucleus would blow itself apart. Electrons are held in position by the electrostatic attraction between the positively charged protons and negatively charged electrons. If enough energy is supplied, this attraction can be overcome and electrons can be removed from the atom. In this section you will learn how to calculate this energy, known as the ionisation energy.

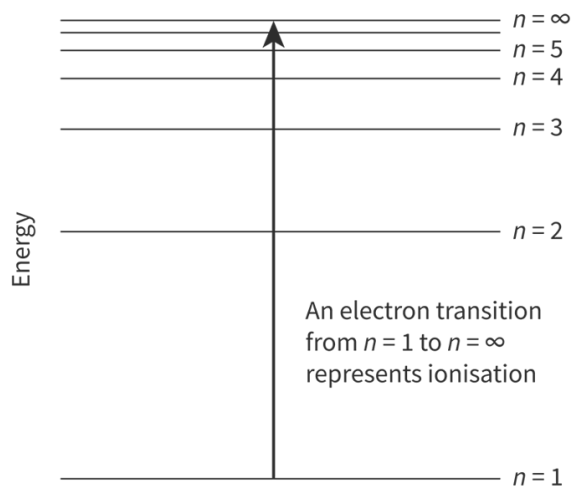
### Making connections

Ionisation energy is covered in more detail in section S3.1.3b (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/>).

The periodic table places elements in rows and columns, organising the elements based on similarities. There are 18 vertical columns, known as groups, and 7 horizontal rows, known as periods. Ionisation energy decreases for elements that are in the same group and increases for elements in the same period.

## Trends in ionisation energy

Ionisation refers to the process of removing electrons from an atom in its ground state. For hydrogen this involves an electron transition from  $n = 1$  to  $n = \infty$  (**Figure 1**). Recall from section S1.3.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/>) that an electron in the  $n = \infty$  energy level is said to be removed from the attraction of the nucleus and the atom has been ionised.



**Figure 1.** When an electron in a hydrogen atom transitions from  $n = 1$  to  $n = \infty$ , it is said to be ionised.

 More information for figure 1

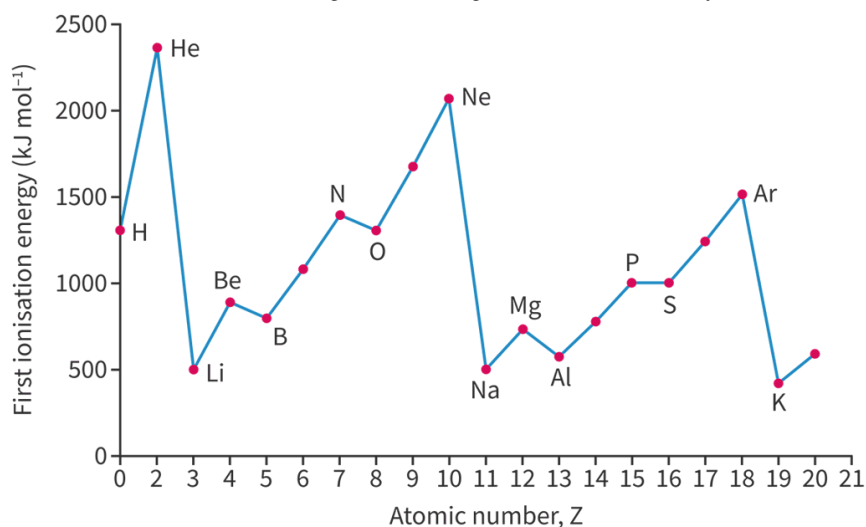
The diagram illustrates an electron transitioning within a hydrogen atom from the  $n=1$  orbit (ground state) to  $n=\infty$  (ionized state). It uses concentric circles to represent the atomic orbits, with labels indicating the  $n=1$ ,  $n=2$ ,  $n=3$ , and  $n=\infty$  energy levels. An arrow points from the  $n=1$  level to  $n=\infty$ , indicating the ionization process where the electron moves from being tightly bound to the nucleus to being completely free. Additional labels may include terms like 'ground state' and 'ionized hydrogen atom.'

[Generated by AI]

To remove electrons from an atom, energy is required to overcome the electrostatic attraction between the positively charged nucleus and the negatively charged electrons. The strength of this attraction depends on two factors:

1. the number of protons in the nucleus, which is known as the nuclear charge.
2. the distance between the nucleus and the outer electrons, which is known as the atomic radius.

The general trend in ionisation energy is shown in the graph in **Figure 2**. This is covered in more detail in [section S3.1.3b](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/>).



**Figure 2.** The trend in ionisation for the first 20 elements in the periodic table.

 More information for figure 2

The graph illustrates the trend in ionization energies for the first 20 elements of the periodic table. The X-axis represents the atomic number, going from 1 (Hydrogen) to 20 (Calcium). The Y-axis indicates the ionization energy in electron volts (eV), with labeled intervals. The graph showcases two series of peaks and troughs. Notable features include:

- A sharp increase in ionization energy from Hydrogen to Helium.
- A dip at Lithium, followed by a rise to Beryllium and another dip at Boron.
- Steady increases and periodic dips are seen again at elements Nitrogen and Oxygen, highlighting the discontinuities between certain elements (e.g., Be to B and N to O).
- For period 3, a rise from Sodium, with dips and rises similar to the pattern in period 2.
- The graph highlights elements with higher ionization energies and discontinuities that are less pronounced between elements 11 to 18 (Sodium to Argon), showing similar patterns to the previous period but shifted to higher values on the Y-axis.
- Concludes with a discussion of periodic trends and exceptions to these trends in the context of periods.

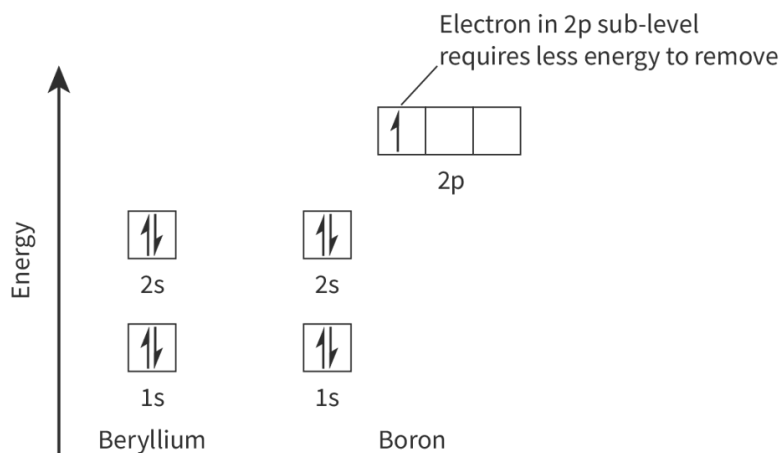
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Looking at **Figure 2**, what overall trend can you observe for elements 3 to 10 (lithium to neon)? Are there any elements that don't seem to follow the trend? What about for elements 11 to 18 (sodium to argon)? Is the general trend the same? Once again, are there any elements that don't seem to follow the trend? Close inspection of this graph reveals two discontinuities in each period – between Be and B and also N and O in period 2 and between Mg and Al and also P and S in period 3.

The decrease in first ionisation energy from beryllium to boron can be explained by examining their electron configurations:

- beryllium has the electron configuration  $1s^2 2s^2$
- boron has the electron configuration  $1s^2 2s^2 2p^1$

Electrons in p orbitals are of higher energy and further from the nucleus than electrons in s orbitals, therefore they require less energy to remove (**Figure 3**).



**Figure 3.** The orbital diagrams for beryllium and boron.

 More information for figure 3

This image displays orbital diagrams for beryllium and boron, comparing their energy levels. On the vertical axis, energy is labeled, indicating increasing energy as you move upwards. The diagram for beryllium shows two energy levels, labeled as 1s and 2s, each with a filled orbital represented by arrows indicating electron spin within a rectangular box.

Next to beryllium, the diagram for boron also shows the 1s and 2s levels with filled orbitals. Above these, a 2p level is shown with one single electron indicated by an arrow in a partially filled rectangular box. An annotation reads, "Electron in 2p sub-level requires less energy to remove," emphasizing that the 2p orbital's electron is of higher energy, thus easier to remove than those in s orbitals.

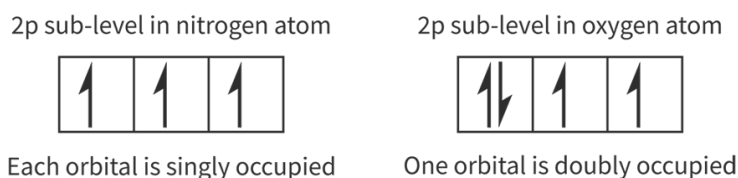
[Generated by AI]

The same explanation can be applied for the decrease in ionisation energy from magnesium to aluminium, except that the electron configurations are  $1s^2 2s^2 2p^6 3s^2$  and  $1s^2 2s^2 2p^6 3s^2 3p^1$  respectively. The electron removed from aluminium is located in a 3p orbital which is further from the nucleus and requires less energy to remove than the electron removed from the 3s orbital in magnesium.

To explain the decrease in ionisation energy between nitrogen and oxygen we will examine their electron configurations:

- nitrogen has the electron configuration  $1s^2 2s^2 2p^3$
- oxygen has the electron configuration  $1s^2 2s^2 2p^4$

**Figure 4** shows the electron in box diagrams for nitrogen and oxygen.



**Figure 4.** The orbital diagrams for nitrogen and oxygen.

 More information for figure 4

The image depicts orbital diagrams for nitrogen and oxygen. In the nitrogen diagram, there are three boxes representing p orbitals, each containing a single upward arrow, indicating singly occupied orbitals. In contrast, the oxygen diagram shows three boxes as well, but one of the boxes contains two arrows (one up and one down), indicating a doubly occupied orbital. This illustrates how nitrogen has three singly occupied p orbitals, while oxygen has one doubly occupied orbital, aligning with the description that an electron in the doubly occupied orbital of oxygen requires less energy to remove due to repulsion from the second electron.

[Generated by AI]

Nitrogen has three singly occupied p orbitals but in oxygen, one orbital is doubly occupied. An electron in a doubly occupied orbital is repelled by the second electron and requires less energy to remove than an electron in a half-filled orbital. The same explanation applies for the decrease in ionisation energy from phosphorus to sulfur, except that the electrons are being removed from the 3p sublevel.

## Calculating ionisation energy

In section S1.3.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>), you learned that energy levels converge at high energy. The point at which the energy levels in an atom converge is known as the convergence limit. At this point the electron has been removed from the attraction of the nucleus and the atom has been ionised. If the wavelength or frequency of the convergence limit is known, the ionisation energy can be calculated. To calculate the ionisation energy, we will make use of the equations and constants in **Tables 1** and **2**.

**Table 1. Equation for energy.**

| Equation | Symbols                                             | Units                                    |
|----------|-----------------------------------------------------|------------------------------------------|
| $E = hf$ | $E$ is energy                                       | Joules (J)                               |
|          | $h$ is Planck's constant,<br>$6.63 \times 10^{-34}$ | Joule seconds (J s)                      |
|          | $f$ is the frequency                                | seconds <sup>-1</sup> (s <sup>-1</sup> ) |

**Table 2. Equation for speed of light.**

| Equation        | Symbols                                         | Units                                  |
|-----------------|-------------------------------------------------|----------------------------------------|
| $c = \lambda f$ | $c$ is the speed of light<br>$3.00 \times 10^8$ | metres per second (m s <sup>-1</sup> ) |
|                 | $\lambda$ is wavelength                         | metres (m)                             |

| Equation | Symbols              | Units                                    |
|----------|----------------------|------------------------------------------|
|          | $f$ is the frequency | seconds <sup>-1</sup> (s <sup>-1</sup> ) |

### Study skills

The equations and constants listed above can be found in [section 1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/>) and [section 2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/>) of the DP chemistry data booklet.

### Worked example 1

The frequency of the convergence limit for the hydrogen atom is  $3.28 \times 10^{15} \text{ s}^{-1}$ .

Calculate the ionisation energy for hydrogen in  $\text{kJ mol}^{-1}$ .

In the question the frequency of convergence limit is given so we can use the equation:

$$\begin{aligned} E &= hf \\ &= 6.63 \times 10^{-34} \times 3.28 \times 10^{15} \\ &= 2.17 \times 10^{-18} \text{ J} \end{aligned}$$

This is the ionisation energy in joules (J) for a single atom of hydrogen. To calculate the ionisation energy in  $\text{kJ mol}^{-1}$ , we need to multiply by the Avogadro constant and divide by 1000 to convert to kJ.

$$\begin{aligned} E &= \frac{2.17 \times 10^{-18} \times 6.02 \times 10^{23}}{1000} \\ &= 1310 \text{ kJ mol}^{-1} \text{ (3 s.f.)} \end{aligned}$$

### Worked example 2

Determine the wavelength of a photon that will cause the first ionisation of copper. The ionisation energy of copper is  $745 \text{ kJ mol}^{-1}$ .

In the question we will calculate the wavelength of a single photon of energy required to ionise a copper atom.

First, we will find the energy of a photon in joules (J) to ionise a single copper atom:

$$\begin{aligned} E &= \frac{745 \times 1000}{6.02 \times 10^{23}} \\ &= 1.24 \times 10^{-18} \text{ J (3 s.f.)} \end{aligned}$$

Next, we will calculate the frequency of the photon:

$$E = hf$$

$$1.24 \times 10^{-18} = 6.63 \times 10^{-34} \times f$$

$$f = 1.87 \times 10^{15} \text{ s}^{-1} \text{ (3 s.f.)}$$

Finally, we will calculate the wavelength of the photon:

$$c = \lambda f$$

$$3.00 \times 10^8 = 1.87 \times 10^{15} \times \lambda$$

$$\lambda = 1.60 \times 10^{-7} \text{ m (3 s.f.)}$$

## Theory of Knowledge

Observing colour with our sense of sight is something not all people can do, and many 'colours' fall out of the visible spectrum. To what extent do we put faith in technology to 'see' what we cannot see ourselves?

## Making connections

How does the trend in ionisation energy values across a period and down a group explain the trends in properties of metals and non-metals? In subtopic S3.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>), you will learn about the trend in ionisation energy can be used to distinguish between metal and non-metal elements.

In the following activity, you will practise how to calculate the ionisation energy using a database.

## Activity

- **IB learner profile attribute:** Inquirers
- **Approaches to learning:** Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

In this activity, you will use a database to calculate the ionisation energy from the convergence limit of an atom.

1. Start by clicking on this link.

<http://physics.nist.gov/PhysRefData/Handbook/periodictable.htm>   
(<http://physics.nist.gov/PhysRefData/Handbook/periodictable.htm>)

You should see this periodic table below. Choose an element from the periodic table and click on it. In this example, we will choose carbon.

Select an element to access data

|               |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|---------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 1H            |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | 2He  |
| 3Li           | 4Be  |      |      |      |      |      |      |      |      |      |      | 5B   | 6C   | 7N   | 8O   | 9F   | 10Ne |
| 11Na          | 12Mg |      |      |      |      |      |      |      |      |      |      | 13Al | 14Si | 15P  | 16S  | 17Cl | 18Ar |
| 19K           | 20Ca | 21Sc | 22Ti | 23V  | 24Cr | 25Mn | 26Fe | 27Co | 28Ni | 29Cu | 30Zn | 31Ga | 32Ge | 33As | 34Se | 35Br | 36Kr |
| 37Rb          | 38Sr | 39Y  | 40Zr | 41Nb | 42Mo | 43Tc | 44Ru | 45Rh | 46Pd | 47Ag | 48Cd | 49In | 50Sn | 51Sb | 52Te | 53I  | 54Xe |
| 55Cs          | 56Ba | *    | 72Hf | 73Ta | 74W  | 75Re | 76Os | 77Ir | 78Pt | 79Au | 80Hg | 81Tl | 82Pb | 83Bi | 84Po | 85At | 86Rn |
| 87Fr          | 88Ra | +    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| * Lanthanides |      |      | 57La | 58Ce | 59Pr | 60Nd | 61Pm | 62Sm | 63Eu | 64Gd | 65Tb | 66Dy | 67Ho | 68Er | 69Tm | 70Yb | 71Lu |
| + Actinides   |      |      | 89Ac | 90Th | 91Pa | 92U  | 93Np | 94Pu | 95Am | 96Cm | 97Bk | 98Cf | 99Es |      |      |      |      |

 More information

The image shows a periodic table from NIST (National Institute of Standards and Technology). The title suggests selecting an element to access data. The table consists of various chemical elements arranged by atomic number, symbol, and name. Key sections include groups like Lanthanides and Actinides, displayed in separate rows below the main table. Elements are arranged in columns and rows representing groups and periods. Options like "Main Page," "Finding List," "Element Name," and "Atomic Number" are located at the top along with a button to switch to an ASCII version of the table.

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1. Click on 'Neutral atom' and 'Energy levels'.

Carbon (C)

|                         |              |              |               |                |             |              |                  |                |      |
|-------------------------|--------------|--------------|---------------|----------------|-------------|--------------|------------------|----------------|------|
| Other Elements          |              |              |               | Neutral Atom   |             |              |                  | Singly Ionized |      |
| <b>Main Page</b>        | Finding List | Element Name | Atomic Number | Periodic Table | Atomic Data | Strong Lines | Persistent Lines | Energy Levels  | Ref. |
| Switch to ASCII Version |              |              |               |                |             |              |                  |                |      |

 More information

This image is a screenshot of a webpage header featuring navigation buttons related to the element Carbon (C). It has multiple labeled sections. Starting from the left, there is a button labeled 'Main Page' with an option below to 'Switch to ASCII Version.' Next, there are labeled sections titled 'Other Elements' and 'Neutral Atom.' Under 'Other Elements,' the tabs include 'Finding List,' 'Element Name,' 'Atomic Number,' and 'Periodic Table,' each with its own distinct color background. Under 'Neutral Atom,' the tabs are 'Atomic Data,' 'Strong Lines,' 'Persistent Lines,' and 'Energy Levels.' The section labeled 'Singly Ionized' includes 'Persistent Lines,' 'Energy Levels,' and 'Ref.' Each tab is presented in a colored box, indicating distinct navigational links.

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1. Scroll to the bottom of the page where it says 'C I Ground State'. Make a note of the value for the ionisation energy.
2. This value is the wavenumber (in  $\text{cm}^{-1}$ ) of the convergence limit for a carbon atom. To convert to wavelength in metres, we need to take the reciprocal of the wavenumber and divide by 100. So for carbon:

$$\lambda = \frac{1}{\frac{90\,820}{100}}$$

$$= 1.10 \times 10^{-7} \text{ m (3 s.f.)}$$

3. Now that we have the wavelength of the convergence limit of a carbon atom, we can calculate the frequency of the convergence limit.

$$c = \lambda f$$

$$3.00 \times 10^8 = 1.10 \times 10^{-7} \times f$$

$$f = 2.73 \times 10^{15} \text{ s}^{-1}$$

4. Next, we will calculate the ionisation energy in  $\text{kJ mol}^{-1}$ .

$$E = hf$$

$$= 6.63 \times 10^{-34} \times 2.73 \times 10^{15} \text{ s}^{-1}$$

$$= 1.81 \times 10^{-18} \text{ J}$$

This is the ionisation energy in joules (J) for a single atom of carbon. To convert to  $\text{kJ mol}^{-1}$ , we need to multiply by the Avogadro constant and divide by 1000.

$$\frac{1.81 \times 10^{-18} \times 6.02 \times 10^{23}}{1000} = 1090 \text{ kJ mol}^{-1}$$

If we compare this to the value in [section 9](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet, which is  $1086 \text{ kJ mol}^{-1}$ , we can see our calculated value is accurate to three significant figures.

Next, choose a different atom from the periodic table and calculate the ionisation energy in  $\text{kJ mol}^{-1}$ . Once you have calculated a value, check it with [section 9](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet.

# Successive ionisation energies (HL)

## Higher level (HL)

### Learning outcomes

By the end of this section you should be able to deduce the group of an element from its successive ionisation data.

In the previous [section S1.3.6](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/>) you learned that electrons can be removed from an atom and that the energy required to do this is the ionisation energy. But is it possible to remove more than one electron if enough energy is supplied? Would doing so require the same amount of energy or more energy? Rather than just removing one electron, it is possible to continue to remove electrons as long as energy is supplied. These are known as successive ionisation energies and as you will learn in this section, we can use successive ionisation data to determine to which group an element belongs.

## Successive ionisation energies

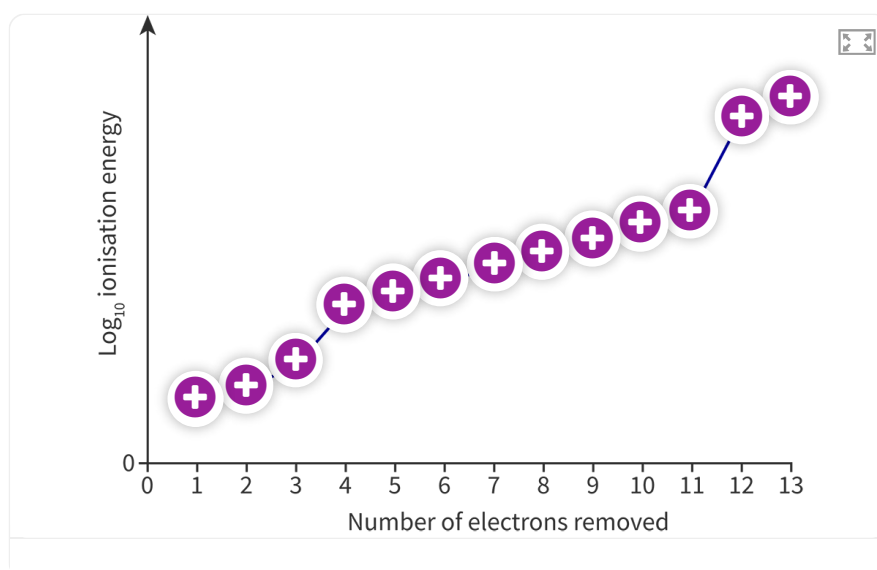
Successive ionisation refers to the process of removing successive electrons from an atom or positive ion. Taking aluminium as an example, we can write equations that represent the successive ionisation energies (**Table 1**). Note that after the first ionisation energy, each successive ionisation removes an electron from an increasingly positive ion. This has an important effect on the value of the ionisation energy and will be discussed later in the section.

**Table 1.** Equations for the successive ionisation of aluminium.

| Ionisation | Equation                                                                     |
|------------|------------------------------------------------------------------------------|
| 1st        | $\text{Al(g)} \rightarrow \text{Al}^+(\text{g}) + \text{e}^-$                |
| 2nd        | $\text{Al}^+(\text{g}) \rightarrow \text{Al}^{2+}(\text{g}) + \text{e}^-$    |
| 3rd        | $\text{Al}^{2+}(\text{g}) \rightarrow \text{Al}^{3+}(\text{g}) + \text{e}^-$ |
| 4th        | $\text{Al}^{3+}(\text{g}) \rightarrow \text{Al}^{4+}(\text{g}) + \text{e}^-$ |
| 5th        | $\text{Al}^{4+}(\text{g}) \rightarrow \text{Al}^{5+}(\text{g}) + \text{e}^-$ |
| 6th        | $\text{Al}^{5+}(\text{g}) \rightarrow \text{Al}^{6+}(\text{g}) + \text{e}^-$ |

| Ionisation | Equation                                                                         |
|------------|----------------------------------------------------------------------------------|
| 7th        | $\text{Al}^{6+}(\text{g}) \rightarrow \text{Al}^{7+}(\text{g}) + \text{e}^{-}$   |
| 8th        | $\text{Al}^{7+}(\text{g}) \rightarrow \text{Al}^{8+}(\text{g}) + \text{e}^{-}$   |
| 9th        | $\text{Al}^{8+}(\text{g}) \rightarrow \text{Al}^{9+}(\text{g}) + \text{e}^{-}$   |
| 10th       | $\text{Al}^{9+}(\text{g}) \rightarrow \text{Al}^{10+}(\text{g}) + \text{e}^{-}$  |
| 11th       | $\text{Al}^{10+}(\text{g}) \rightarrow \text{Al}^{11+}(\text{g}) + \text{e}^{-}$ |
| 12th       | $\text{Al}^{11+}(\text{g}) \rightarrow \text{Al}^{12+}(\text{g}) + \text{e}^{-}$ |
| 13th       | $\text{Al}^{12+}(\text{g}) \rightarrow \text{Al}^{13+}(\text{g}) + \text{e}^{-}$ |

**Interactive 1** shows a graph of the successive ionisation energies of aluminium. Move your mouse over each point to see from which sublevel the electron is being removed. From which sublevel do we start removing electrons? Is it from the sublevel closest to the nucleus or furthest from the nucleus?



### Interactive 1. The Successive Ionisation Energies of Aluminium.

 More information for interactive 1

An interactive graph exhibits the ionisation energies of aluminium. The x-axis represents the number of electrons removed from aluminium, ranging from 0 to 13, while the y-axis shows the log10 of ionisation energy.

The graph features an increasing trend, with plus signs marking interactive hotspots along the line. Selecting these hotspots reveals the subshell from which the electrons are removed.

The number of electrons removed and their corresponding sublevels are listed in a table below:

This graph helps users understand that electrons removed from inner orbitals have higher ionisation energies, while those from outer orbitals require less energy to be removed.

| Number of electrons removed | Sublevel |
|-----------------------------|----------|
| 1                           | 3p       |
| 2                           | 3s       |
| 3                           | 3s       |
| 4                           | 2p       |
| 5                           | 2p       |
| 6                           | 2p       |
| 7                           | 2p       |
| 8                           | 2p       |
| 9                           | 2p       |
| 10                          | 2s       |
| 11                          | 2s       |
| 12                          | 1s       |
| 13                          | 1s       |

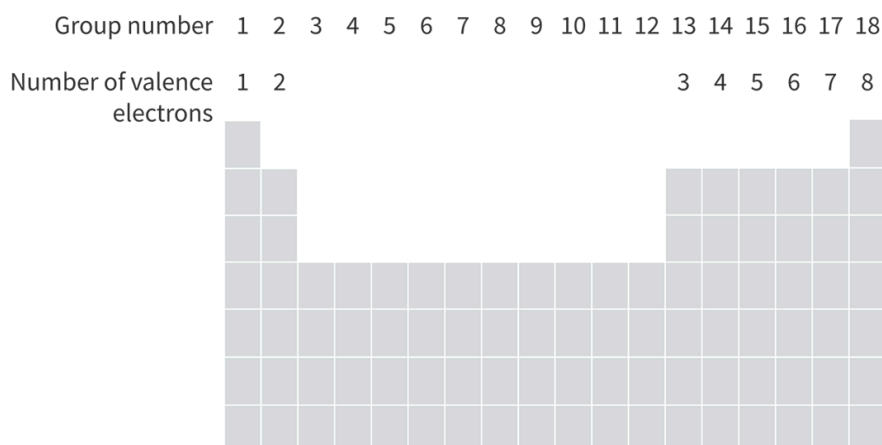
The order in which electrons are removed is from the higher energy sublevels first – so the first electron is removed from the 3p sublevel, and the last electron is removed from the 1s sublevel. When electrons are removed, they are always removed from the highest energy sublevel first.

From the graph we can also see that successive ionisation energies show an increase. The reason for this is that after the first ionisation, the electrons are being removed from an increasingly positive ion. This leads to an increase in the electrostatic attraction between the nucleus and the remaining electrons and an increase in the ionisation energy.

## Making connections

Before you proceed with the rest of this section, it is helpful to learn about the groups of the periodic table; this is covered in [section S3.1.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-the-periodic-table-id-43466/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-the-periodic-table-id-43466/>).

Successive ionisation energies provide evidence as to the electron configuration of an atom, in particular with respect to the number of valence electrons. Recall that valence electrons are those electrons located in the outermost shell of an atom. Elements with the same number of valence electrons are placed in the same group on the periodic table. The periodic table is divided into vertical groups numbered from 1–18. The main group elements are those located in groups 1 and 2 and 13 to 18. For these elements, **Figure 1** tells us how many valence electrons an atom has.



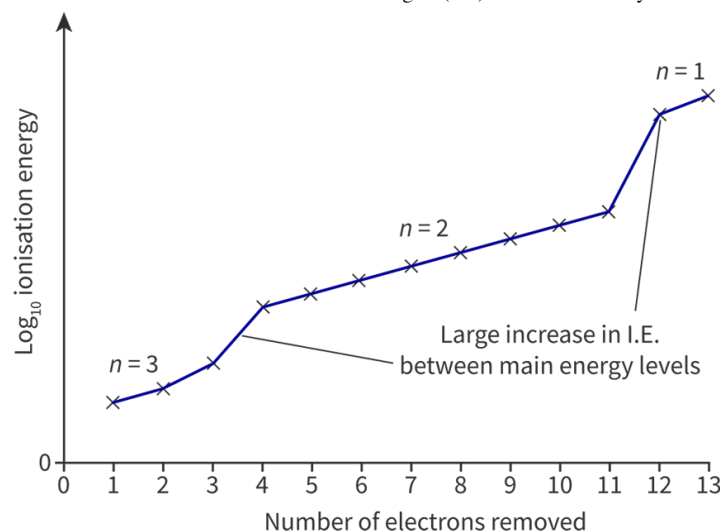
**Figure 1.** The number of valence electrons in each main group of the periodic table.

 More information for figure 1

The image shows a periodic table of elements, highlighting the main groups which indicate the number of valence electrons. The main groups are marked as vertical columns, and the applicable groups range from 1 to 2 and 13 to 18, which are shaded differently to show their significance. This differentiation helps in understanding the electron configuration by emphasizing the vertical arrangement of elements having the same number of valence electrons. Light grey squares represent the elements in the table, with darker shading for the main groups. The periodic table is structured with elements arranged in order of increasing atomic number.

[Generated by AI]

**Figure 2** shows the successive ionisation energies of aluminium again, but this time showing the main energy level to which each electron belongs. What do you notice about the graph? Is the increase in ionisation energy constant or are there large increases every so often? And what do these large increases mean?



**Figure 2.** The successive ionisation energies of aluminium.

More information for figure 2

The graph illustrates the successive ionisation energies of aluminium. The X-axis represents the electron number removed, ranging from 1 to approximately 13, while the Y-axis represents the ionisation energy in arbitrary units. The graph shows a series of step-like increases with two significant jumps in energy levels. The first major increase occurs after the removal of the third electron, while the second substantial rise happens after the removal of the eleventh electron. These large increases indicate electrons being removed from different energy levels, with the 4th and 12th electrons notably requiring more energy as they belong to lower energy levels than previous ones. This pattern supports the concept of energy levels within the atom.

[Generated by AI]

Two large increases can be seen in the graph; one after the 3rd electron is removed and another after the 11th electron is removed. These large increases correspond to the 4th and 12th electrons being removed from different energy levels to the previous electrons. The 3rd electron is removed from  $n = 3$  whereas the 4th electron is removed from  $n = 2$ . And the 11th electron is removed from  $n = 2$  whereas the 12th electron is removed from  $n = 1$ . These large increases are further evidence for the existence of energy levels in the atom.

As mentioned earlier we can also use these large increases to determine to which group an element belongs, based on the number of valence electrons an atom has. The first large increase in a graph of successive ionisation energies occurs after the removal of the valence electrons. To determine the group that the element belongs to, we need to count the number of electrons before this large increase, which tells us the number of valence electrons in the atom and to which group the element belongs. So for aluminium this occurs after the 3rd electron is removed which tells us that aluminium has three valence electrons and is located in group 13 of the periodic table.

Data for successive ionisation energies can also be represented in table form. **Table 2** shows the successive ionisation data for the first four ionisations of aluminium.

**Table 2.** Successive ionisation data for the first four ionisations of aluminium.

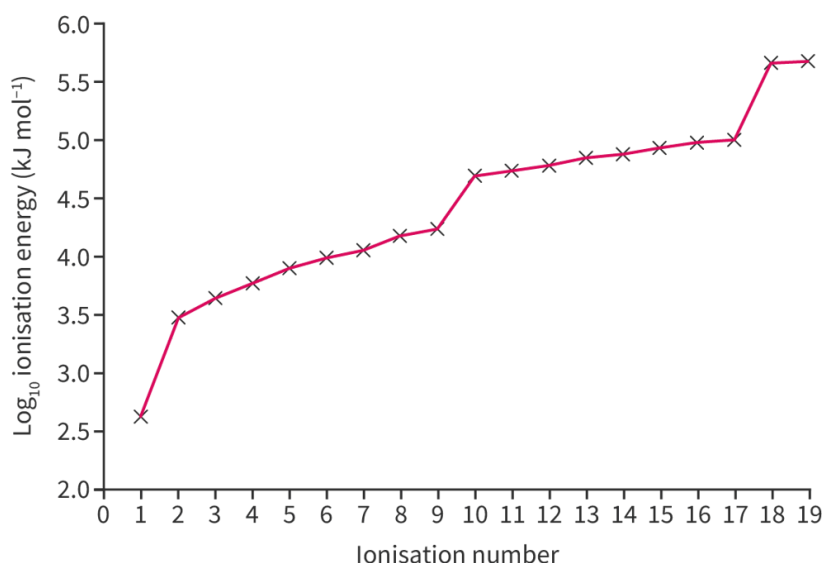
| Ionisation                               | 1st | 2nd  | 3rd  | 4th   |
|------------------------------------------|-----|------|------|-------|
| Ionisation energy / $\text{kJ mol}^{-1}$ | 577 | 1820 | 2740 | 11500 |

Once again, we see the large increase in ionisation energy occurs after the third electron is removed, indicating that the fourth electron is removed from an energy level closer to the nucleus. The large increase after the third electron is removed tells us once again that aluminium is located in group 13 of the periodic table.

## Worked example 1

**Figure 3** shows all the successive ionisation energies of an element.

- How many occupied energy levels does the atom contain?
- From which energy level is the first electron removed?
- To which group of the periodic table does the element belong?
- What is the identity of the element?



**Figure 3.** Successive ionisation energies of an element.

More information for figure 3

The graph displays the successive ionisation energies of an element on a logarithmic scale. The X-axis represents the ionisation number, ranging from 0 to 19. The Y-axis shows the log base 10 of ionisation energy in  $\text{kJ mol}^{-1}$ , ranging from 2.0 to 6.0. Data points are marked with an 'X' and connected by a line. The graph indicates an overall upward trend, with slight variations in slope at various points. Notably, there are more pronounced increases in ionisation energy around the ionisation numbers 9 to 10, and 17 to 18, suggesting higher energy requirements to remove these electrons. Between these points, the increases are more gradual, indicating less energy is needed to remove those electrons.

[Generated by AI]

- The number of occupied energy levels can be determined by the number of 'jumps' or large increases in ionisation energy. There are three large increases in ionisation energy so there are four occupied energy levels.

(b) The first electron is removed from the  $n = 4$  energy level.

(c) There is one electron removed before the first big increase in ionisation energy, so the element has one valence electron and is located in group one of the periodic table.

(d) There are a total of 19 electrons removed so the atom must have 19 protons. The element with 19 protons is potassium ( $Z = 19$ ).

## Worked example 2

Data for the successive ionisation energies of an unknown element is shown in **Table 3**. Deduce how many valence electrons the unknown element has and to which group of the periodic table it belongs.

**Table 3.** Successive ionisation data for an unknown element.

| Ionisation                               | 1st | 2nd  | 3rd  |   |
|------------------------------------------|-----|------|------|---|
| Ionisation energy / $\text{kJ mol}^{-1}$ | 590 | 1145 | 4912 | 6 |

To determine the number of valence electrons and group number, we look for the large increase in ionisation energy. The large increase occurs after the second electron is removed. The ionisation energy increases from  $1145 \text{ kJ mol}^{-1}$  to  $4912 \text{ kJ mol}^{-1}$  because the third electron is removed from an energy level closer to the nucleus. Therefore, the element has two valence electrons and is located in group 2 of the periodic table.

### Making connections

Subtopic S3.1 (HL) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>) covers how patterns of successive ionisation energies of transition elements help to explain the variable oxidation states of these elements?

In this activity, you will plot a graph of successive ionisation energies and explain to which group the element belongs.

### Activity

- **IB learner profile attribute:** Inquirers
- **Approaches to learning:** Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity



In this activity, you will choose an element and plot a graph of its successive ionisation energies.

1. Click on the link below.

<https://www.webelements.com/atoms.html>   
(<https://www.webelements.com/atoms.html>)

2. Choose an element from group 15—18 in period three of the periodic table and click on that element.
3. Write the full electron configuration of your chosen element.
4. Find the successive ionisation data for your chosen element and record it in table form.
5. Use a spreadsheet such as Google Sheets to plot a graph of the number of electrons removed (on the x-axis) against ionisation energy (on the y-axis).
6. Label the electrons with the main energy sublevel they belong to. Label the large increases in ionisation energy and explain each increase.
7. Explain how the graph confirms to which group your chosen element belongs.